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Public Procurement and Supplier Job Creation: Insights from Auctions

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ABSTRACT

Public procurement contracts (PPCs) of goods, services, and works are about one-tenth of the global gross domestic product. Much research has been conducted on government spending and its aggregate effects, but the evidence is scarce at the micro-level. We exploit sealed-bid PPC auctions of construction works, discontinuity in bidders' win margin, and firms' daily employment variation to provide a causal estimate of winning a PPC on firms' employment. Winning a PPC has a small positive impact on a firm's short-run employment. We investigate mechanisms and heterogeneity that can explain the small initial magnitudes. We find no compelling evidence in favor of political connections, an information leakage channel, or PPC size as explanations for the small magnitude. Our investigation of a longer period shows that the impact phases out in less than a year. The lack of a long-term impact is due to runners-up winning more PPCs and runners-up substituting for more market revenue in the year after closely losing a PPC. Finally, the impacts are concentrated in construction firms that conduct most contracted work *in-house*. The final estimation shows the effect is 3.7 new employees per PPC with a public cost per job created at €58,600 (€49,800–€65,100).

JEL CLASSIFICATIONS: H57; D44; D22; M51

1. INTRODUCTION

The World Bank estimates that public purchases of goods, services, and works (i.e., public procurement contracts, PPC) amount to 12% of the world's gross domestic product (GDP).[1] Macro-research shows that government purchase creates fiscal multipliers and thus fosters economic activity (for a review, see Ramey 2011), with additional evidence showing it can also spur innovation (i.e., critical components of iPhone, Mazzucato 2011) and contribute to achieving social outcomes (i.e., via green procurement, Testa et al. 2016).

1 Link: https://blogs.worldbank.org/developmenttalk/how-large-public-procurement (accessed February 15, 2021).

However, there is little micro evidence on how firms behave when faced with a public demand shock (Gugler et al. 2020; Ferraz et al. 2021).[2] We speak to this scarce literature by estimating the effect of PPC on firms' job creation, and in doing so, to our knowledge, we are first to examine the "*winner-runner up*" identification strategy with respect to the political networks and information leakage channel (Szucs 2017; Baltrunaite 2020). We are also the first to estimate the heterogeneous effects of PPC concerning worker characteristics and employment history, the dose–response functions of PPC sizes, and the heterogeneity of PPC effects concerning firms' outsourcing decisions.

Given a steady market revenue, receiving an additional PPC represents an increase in public revenue and, therefore, the firm's total revenue. Standard short-run product function logic suggests that an increase in firms' output leads to an increase in labor inputs (Hart 2017).[3] The theory has recently accumulated empirical support for the positive effect of PPC on employment (Gugler et al. 2020; Ferraz et al. 2021).[4] It is also possible that the effect of PPC on firms' labor inputs is larger than the potential effect of similar contracts in the private market, as several authors suggest PPCs can be more profitable and less risky (Duggan et al. 2016; Fadic 2019). However, the co-movement of output and labor is not always observed in the data (Hamermesh 1989). Co-movement between negative output change and labor adjustment was not found by Hamermesh (1989), while Gugler et al. (2020) do not find co-movement of positive output change and labor adjustment during a recession. Labor hoarding is the main theoretical reason for no co-movement of output and labor (Bertola and Caballero 1994). When a profit-maximizing firm does not acquire a PPC, assuming fully utilized capacity and constant or increasing demand, it might still be tempted not to immediately fire its workers due to future hires' hiring and training costs. When a firm does acquire a PPC, the new PPC might not be large enough for firms to increase their employment. For example, the firm has economic slack and utilizes unexploited labor to produce newly demanded output, or it reallocates workers from recently finished projects to the new PPC (Fadic 2019).

We merge eight census datasets to provide several novel findings on the PPC effect on firms' employment. Allocation of PPCs is not random, so identifying the impact of PPC on firm employment faces several endogeneity issues. Recent micro-level studies suggest an auction-based identification strategy to solve these issues (i.e., Gugler et al. 2020; Ferraz et al. 2021). Ferraz et al. (2021) use instrumental variables approach exploiting the share of close auctions won, while Gugler et al. (2020) use discontinuity in close auctions in a difference-in-differences setting to compare the employment growth of auction winners with those of runners-up in the weeks after the auction results. Similar to the previous two studies, we exploit exogenous variation in a sealed bid auction win margin and, critically, combine it with employer–employee daily hiring and firing data to estimate the effect on jobs in the weeks and months after winning the PPC. Unlike the previous two studies, which do not address political economy issues, we examine the allocation of PPC concerning unique political and business network data. We collect names and surnames of all firms' owners, chief executive officers (CEOs), supervisory board members, the register of firm donations to political parties, and names and surnames of about 30,000 politicians at the national, regional, and

[2] We use the words "*firms*" and "*suppliers*" interchangeably.

[3] In other words, we are theoretically embedded in the firm's short-run production theory, where capital is fixed, and labor is a variable factor. We are also embedded in the market for inputs and the theory on firms' short-run labor demand. A firm's decision on hiring is based, on the one hand, on the marginal cost of labor (MC_L), and on the other hand, on the marginal revenue product of labor (MRP_L) which is a product of the marginal product of labor (MP_L) and firms' marginal revenue of the final product (MR_A).

[4] There are also interesting studies quantifying the connection between receiving (i) a standard PPCs or newer "PPCs for innovation" with (ii) firm growth and innovation, but these studies do not have a source of exogenous variation nor discuss public costs per job created (e.g., Stojčić et al. 2020).

local levels. We develop multiple first- and second-order political connection proxies to show several interesting findings. Most importantly, winners and runners-up in close auctions are, on average, similarly politically connected.

We show a positive impact on firms' short-run employment ten weeks after being awarded a PPC. This finding is in line with the previous studies in the short-run (during expansion, Gugler et al. 2020; Ferraz et al. 2021), but our results show a different magnitude with the PPC effect at 2.7 employees. The initial estimate of the cost per job is high at €93,223, considerably higher than the estimates in Austria (€22,246 in Gugler et al. 2020) and Brazil ($58,807 in Ferraz et al. 2021), but smaller than estimates with more aggregate data in the United States ($125,000 in Wilson 2012).[5] Studies investigating the effects of public grants on firm employment find a cheaper public cost per job than those reported in this study (among others Criscuolo et al. 2019; Dvouletý et al. 2021; Srhoj and Zilic 2021). The anatomy of identified new hires shows that the majority are male, hired on fixed-term contracts, and on job positions requiring lower education. Additionally, most new hires are working in the country for the first time, from which a non-negotiable share is probably foreign workers. Previously employed new hires usually transfer from other construction firms or are experiencing a brief unemployment spell.

Our results also explain the small magnitude of the PPC effect on firms' employment. Runners-up in close auctions start working more in the private market, thus indicating substitution of the public for the private market. Relatedly, we find suggestive evidence against the Matthew effect, which departs from previous studies (see Ferraz et al. 2021). The Matthew effect also termed the Matthew effect of accumulated advantage suggests that the rich get richer and the poor get poorer, which implies that those firms that win a PPC in a close auction get more and more PPCs. However, we find that firms winning a PPC in a close auction receive fewer PPCs in the next year. These two reasons explain why the PPC effect on firms' employment phases out in a year.

In the short run, we provide new findings on the dose–response functions. We construct relative measures of the contract size, dividing the contract size by the firm's yearly revenue. Relatively small contracts, below 7% of the firm's revenue, do not cause a positive employment reaction by the winners. This finding suggests that the designed public demand shock has to be large enough to cause an increase in firms' labor inputs, as opposed to conducting the work with existing capacities. We find the most noticeable effects for contracts above 20% of the firm's yearly revenue. Finally, our study shows new heterogeneity concerning construction of firms' outsourcing intensity and the intensity of using external workers as labor inputs. We show that the effects are concentrated in firms that conduct most of the work in-house. Firms employ about 3.7 new employees per PPC on average, with the public cost per job directly created at €58,600 (€49,800–€65,100).

The rest of the study is as follows. After Section 1, the second section provides details on the institutional setting, including the Republic of Croatia (from now on Croatia), the public procurement market, the formal rules of auctions, and data. The third section explains our identification strategy. The fourth section provides the main results, and the fifth provides results on mechanisms and heterogeneity. The final fifth section concludes.

2. INSTITUTIONAL SETTING

The institutional setting is Croatia. The Great Recession of 2007–9 spilled over to Croatia in 2009, which then experienced a long recession (2009–14). The recession resulted in a

[5] The main comparison message does not change if we convert monetary values from national currencies to purchasing power parity US dollars.

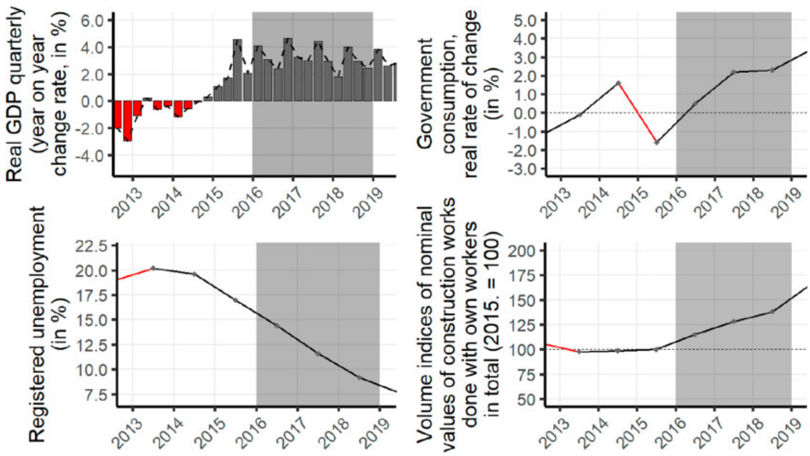

Figure 1. Macroeconomic indicators in Croatia 2013–19.

Source: Croatian National Bank, Macroeconomic Developments and Outlook No. 9 (p. 4); Croatian Statistics Office, Construction—Orders and Completed Works (sheet 3.1.3.).

cumulative GDP drop of 12% and a doubling of the unemployment rate (for policy responses, see Srhoj et al. 2021, 2023; Srhoj and Zilic 2021). The expansion period started in the fourth quarter of 2014, about a year after Croatia entered the European Union (July 2013). This study investigates PPCs in 2016–18, which is interesting for several reasons. Figure 1 shows the real GDP quarter-to-quarter change rate in percentages, showing that after the initial full year of recovery (2015), 2016–18 experienced a stable 4% quarterly GDP growth. Registered unemployment sharply fell from 15% to below 10%, and government consumption sharply increased.

The study focuses on the construction sector, which represents about 4.4% of Croatia's GDP and 6.8% of employment (Buturac 2019). According to the Institute of Economics Zagreb, the value of construction works increased significantly in the analyzed periods: 4.1% from 2016 to 2017, then 9.7% from 2017 to 2018 (Buturac 2018, 2019) after a substantial decline during the recession. Figure 1 also shows a sharp increase in the volume indices of nominal values of construction works from 2016 to 2019 (about 35%). These macroeconomic indicators show that private demand for construction work increased during our analyzed period in the expansion. In other words, we investigate the effect of PPC on construction firms' employment during an expansion period when construction firms have increasingly available private demand.

Finally, according to Transparency International Corruption Perception Index, out of 179 countries, Croatia shares 63rd position with Cuba, São Tomé and Príncipe, and Belarus.[6] Thus, it does not come as a surprise that Croatia's economic system is sometimes described as crony capitalism, where firms do not thrive based on innovation and risk-taking, but on political connections and corruption (among others, see Vuković 2020; Gnip 2022). For the reason above, Croatia is an ideal laboratory for applying and examining the auction-based identification strategy concerning the potential political economy confounders in identifying the effect of PPC on firms' employment.

[6] Link: https://www.transparency.org/en/cpi/2020 (accessed March 2, 2021). A brief literature review on political connections and public procurement contracts can be found in Appendix A.

2.1 Institutional details on public procurement

All PPCs with an estimated value above 200,000 kuna (approx. €27,000) are awarded to suppliers based on competitive tendering via online auctions administered by the Official Gazette.[7] The award process is as follows. The contracting authority publishes the tender via the Official Gazette's online registry of public procurement (i.e., EOJN).[8] Firms (suppliers) are informed of the tender via the EOJN and decide whether to bid or not. Interested firms have to place their bid before the pre-specified deadline, after which firms are no longer allowed to place bids. Auctions are sealed bid first offer. Thus, bidders do not get information on each other's bids, and the best offer wins the PPC. Upon the deadline, the contracting authority opens the bids and awards the PPC to the supplier with the best offer (bid).

Best-offer (bid) criteria are pre-specified at the start of the auction with the supplier selection method. During the period 2011–16, the public procurement market was governed by the Public Procurement Law (NN 90/2011, 143/2013, 83/2013, 13/2014), while from the second half of 2017, it was governed by the new Public Procurement Law (NN 120/16). The required supplier selection method of the Public Procurement Law (NN 90/2011, 143/2013, 83/2013, 13/2014) was the lowest price (LP), and thus the private supplier wins if it submits a valid LP bid.[9] The next Public Procurement Law (NN 120/16) introduced a new mandatory supplier selection method—the most economically advantageous tender (MEAT), which meant simple LP criteria was formally no longer used.[10] The MEAT tender implies that the contracting authority selects the supplier based on a score composed of points gathered based on the price and other quality criteria at the procurer's discretion, such as time of delivery, warranty, or quality. The Law (NN 120/16) introduced the 90% cap on the highest possible weight to be placed on price.[11] We later discuss this change in the supplier selection method but provide evidence that while MEAT was introduced, the contracting authority mostly used 90% weight for the price, which made MEAT very similar to the LP tender. Once a supplier wins a construction PPC, under the condition of fulfilling all PPC obligations, it is at the firm's disposal to decide how the contracted works will be carried out (i.e., in-house, outsourcing, or a mix).

The PPC tenders are supervised by the State Commission for Supervision of Public Procurement Procedures (DKOM).[12] Auction participants with a legal interest in winning the PPC and who believe they have suffered damage because of alleged violations of subjective rights (e.g., favoritism, inside information) have a ten-day window to initiate a dispute before the DKOM by filing a complaint. When deciding on the disputed legal issue, the DKOM acts within the limits of the appellate allegations while it *ex officio* takes into account the procedural requirements and substantive violations listed in the Public Procurement Act. Decisions are announced publicly on the DKOM web site. No appeal is allowed against the decision of the DKOM, but the decision can be challenged in an administrative dispute before the High Administrative Court of the Republic of Croatia.

7 Table A1 (Column 5) shows such contracts represent more than 75% of all PPCs' value. Other columns of Table A1 provide descriptive statistics of PPCs in Croatia from 2015 to 2018.

8 Online registry of public procurement, EOJN, link: https://eojn.nn.hr/ (accessed February 28, 2021).

9 Invalid bids are essentially bids that were turned down by the procurement authority, most frequently because the bidder did not fulfill the minimum criteria (in specifications, capabilities, guarantees, or other aspects), overpriced the contract, faced other law binding circumstances, etc. If a bid is invalid, the procurement authority notes it in the document "*Bid evaluation minutes.*"

10 The new Public Procurement Law (NN 120/16) started on January 1, 2017, but it was not until July 1, 2017, that the application of MEAT became mandatory. There were several exceptions to the rule of not applying exclusively LP, but they are unimportant for our analysis.

11 As shown in Table A1 Column 8, in years 2015 and 2016 more than 97% of contracts were based on LP.

12 Official Gazette, International Treaties, No. 14/01.

2.2 Data

Datasets are collected from several institutions in Croatia and are described in the four subsections.

2.2. 1 Public procurement and auction data

The data on PPC and data on auctions are obtained from the Official Gazette of the Republic of Croatia (2020) and its online registry of public procurement (i.e., EOJN).[13] The PPC data encompass more than 170,000 contracts awarding about 324 billion kuna (€43 billion) with information on contracting authority names and IDs (OIB), their municipal and county IDs, type of PPC, description of the contract, the estimated PPC value, the contracted value of the PPC, the common procurement vocabulary (CPV) codes, the number of bidders per PPC, the name and ID (OIB) of the firm winning the PPC, the municipal and county IDs of the firm, and the dates of auction deadlines and auction results. Public procurement data are enhanced with auction data spanning 2016–18 and contain information at the public procurement bid level: name and ID (OIB) of firms losing in auctions along with their financial offers.[14] The focus of the study is on construction-related PPCs, defined as those in which CPV starts with the number 45, encompassing 800 contracting authorities and 4239 PPCs in the value of €3729 million.

2.2. 2 Employment data

Daily employment data stem from the Croatian Pension Insurance Institute (HZMO) for the period January 2014 until July 2019 and include about 6 million observations (employment spells). On the one hand, HZMO tracks any individual employment process in the Republic of Croatia, and thus, when a person is hired and signs a work contract, they become obligatorily insured in HZMO. On the other hand, when their work contract expires or is terminated, their insurance ceases. For both events, dates are documented. HZMO data are given at the firm–employee level, where firm ID is not anonymous, allowing us to construct variables on employee hires and terminations, their education, age, tenure, gender, and type of current working contract at the firm-day level. In other words, all employment movements within firms and public institutions based in Croatia can be tracked through the HZMO database. Using the work contract and auction dates (Section 2.2.1), it is possible to track employment growth as detailed as in the days before and after auction results.

2.2. 3 Financial data

Firms' financial data stem from the Annual Financial Statements Registry (FINA) and encompass profit and loss statements, balance sheets, and firm demographic information for all limited liability and joint-stock firms in Croatia for the period 2015–19. The FINA dataset is merged with the public procurement auctions data in construction (Section 2.2.1) based on the year and firm ID. This way, it is possible to know each auction participant's financial and firm demographic information (e.g., financial ratios, firm age, municipality, or county of firm headquarters) in 2015–19.

Definitions of variables constructed from the public procurement auction, employment, and firm finance data are provided in Table A2.

[13] Link: https://eojn.nn.hr/ (accessed February 28, 2021). Auction data are web scraped from the EOJN website, and we provide the database to other scientists and students to ignite research in public finance. The data can be accessed via the following website: https://sites.google.com/view/ppc-croatia. We evaluate data quality compared with the Official Annual Statistical Report on Public Procurement 2008–18.

[14] Auction data contain information on the weights used in the MEAT decision criteria (i.e., relative importance of price, quality, or cost) and the total MEAT score of each bidder in the auction.

2.2. 4 Political connections data

Previous data are enriched with several datasets capturing the political economy aspect of public procurement.

1) Court Register for the period 2005–19 provides names and surnames of owners, CEOs, and supervisory board members for each construction firm participating in the public procurement auctions.
2) State Electoral Commission (DIP) provides information on names and surnames of local, regional, and national politicians; their political party; and place of elections (i.e., municipal/city or county-level) covering four local and regional elections during 2005–17 as well as parliament members over the five elections in the period 2003–16.[15] This provides us with 22,083 unique politician full names from which we construct 17,434 unique last name IDs at local (city major, deputies, city/municipality hall members), and 1272 at regional levels (prefect, deputies, county hall members). In addition, 496 unique politicians at the national level (parliament members) are included. The Court Register dataset is merged with politician name lists of DIP in several ways (for details and definitions, see Table A3) to obtain proxies of political connections, which are used to validate the close auctions research design.
3) Gong tool *Mozaik veza*[16] provides information on the names and surnames of politicians and their political parties in Croatia. A total of 6103 unique politicians are in the GONG database. Importantly, our DIP data (2) contain a larger number of politicians than GONG. However, there are as many as 3378 high-ranked politicians in GONG but not in DIP nor parliament data.[17] These politicians' full names are exact-matched with the full names of firm owners, CEOs, and supervisory board members from the Court Register.
4) Eurostat provides information on the geographical latitude and longitude of each municipality in Croatia. Geographical distances from Eurostat are merged with FINA and public procurement data, and then the standard Haversine formula is used to calculate distances between the contracting authority and the construction firm in kilometers. Distance between the two could capture not only transportation costs but also information sharing.[18]
5) Central State Office for Development of Digital Society provides data on 3957 donations by firms and individuals to political parties preceding the parliamentary elections of 2015 and 2016.[19]
6) State Commission for the Control of Public Procurement Procedures (DKOM) provides information on 15,230 official complaints on auctions over the period 2008–18. DKOM provides information on each firm's complaint to each auction and the timing and CPV of each official complaint, regardless of the final court ruling.[20] Intuitively, if

15 Data from the State Electoral Commission is web scraped: https://www.izbori.hr/arhiva-izbora/index.html\#/app/home (accessed March 12, 2021).

16 More information on Gong: https://gong.hr/en/who-are-we/what-is-gong/; more information on the tool *Mozaik veza*: https://www.mozaikveza.hr/ (accessed May 1, 2022).

17 These include individuals such as Ministers or Secretaries of State who were not on the party's list at local, regional, or national elections but are members of a party. With DIP and GONG, the study includes a total of 50,718 observations, but since politicians are present at multiple elections, the data include 25,595 unique politicians.

18 There are 556 municipalities in a population of 4 million in Croatia. The municipality of the firm's headquarters and the municipality of contracting authority are used. Eurostat: https://ec.europa.eu/eurostat/web/gisco/geodata/reference-data/administrative-units-statistical-units/communes\#communes13 (accessed January 20, 2021).

19 These data are web scraped from the following link: https://sredisnjikatalogrh.gov.hr/Politicke-stranke-i-izbori (accessed March 12, 2021).

20 These data are web scraped from the DKOM official website: https://dkom.hr/ (accessed March 12, 2021).

the main results are not driven by political economy issues, they should be robust to excluding the types of construction works and regions with the most complaints. Definitions of political connection variables (proxies) are provided in Table A3. In addition, a graphical illustration of the local-level political connections dummy is provided (in Figure 2).

3. IDENTIFICATION STRATEGY

The study aims to identify the effect of PPC (treatment) on construction firms' employment growth (outcome). Several issues underpin the attempt to determine this causal estimate. Selection into PPC is not random. Winners of PPCs could have an unobservable intent to work on additional projects and have the proper resources, specific expertise, and capabilities to get the work done. The identification strategy applied in this study is the so-called "*winner–runner-up*" identification strategy (also applied in the fields of industrial organization, urban economics, or finance by Greenstone et al. 2010; Kawai 2011; Malmendier et al. 2018; Chassang et al. 2022). Such an identification strategy was applied for estimating the effect of PPC on employment by Gugler et al. (2020).[21] The assumption behind the identification strategy is that auction participants bid competitively. A theoretical framework by Gugler et al. (2015) describes the optimal bidding strategy of an auction participant; it assumes bidder j knows its private cost for PPC project c and provides a bid at an auction. Bidder j knows who other competitors are but does not know their bids. PPC is awarded to the bidder with LP with the condition of not being below the reservation price. The winner of the PPC earns a profit equal to the difference between the bid, and the PPC project cost c, other competitors earn zero profits. Thus, optimal bidders' strategy is a function of PPC project cost, the number of competitors, and their identity such that competitors bid the actual value of the auctioned works minus the shading factor for the estimated bid of other bidders under the condition of the bid being higher than the PPC project costs (see also Maskin and Riley 2000, 2003). In an ideal bidder scenario, the winner would offer a bid higher than the costs of undertaking the PPC, which is just below the bid of the runner-up to maximize its PPC project profits.

If auction participants bid competitively, there could still be selection into winning a PPC because more productive firms can place better bids (i.e., Smith 2017; Gugler et al. 2020; Ferraz et al. 2021). Alternatively, selection might be explained with superb information driven by political connections and favoritism (Baltrunaite 2020; Bosio et al. 2020), which could ambiguously confound the PPC effect downward if such connections can impose high mark-ups for construction works or too low productive firms with substantial organizational slack. Conversely, it could confound the PPC effect upward if the political connections open up more opportunities than just the PPC, for example, unlawful conversion of agricultural to construction land, acquiring permits, and securing favorable regulations. To solve this, this study focuses only on the subsample of winners and runners-up in a particular auction, thereby disregarding bidders placing third or worse. In addition, the study uses only the winners and runners-up in close auctions defined as those with a bid difference below 10%, but it also estimates effects within close auctions defined as those below 8%, 6%, 4%, and auctions between 0.5% and 4%.[22]

[21] Another similar example of the identification strategy is Ferraz et al. (2021).
[22] Win margin in the window between 0.5% and 4% is used to exclude those placing a bid just below the runner-up (below 0.5%) to estimate whether results are driven by these auctions. In the study context, this is to show whether potential information leakage drives the results. If there is information leakage, intuitively, a bid participant with confidential information would

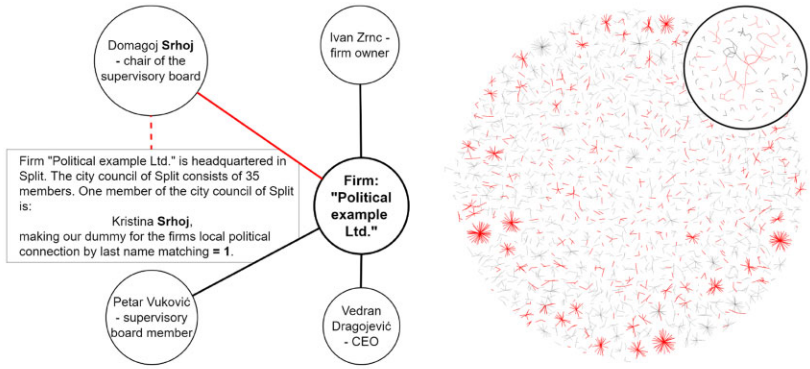

Figure 2. Political connections: an overview.

Notes: The right graph displays the network of politically connected firms (node) and individuals (edge) using the sample of construction firms winning at least one PPC in 2018. The left graphic shows how single-node relations are formed to develop the full network. The red part of the network represents politically connected firms with local connections based on the last name and municipality/city matching (see Table A3). The inner circle shows firms that have common members.

Disclaimer: All names given in the graph are fictional.

Each auction features a date of auction competition results, which is used to define the pre-PPC and post-PPC period. The sample of winners and runners-up is used to construct firms' employment panel data to estimate a *difference-in-differences* model. To construct the dependent variable, we look at the number of employees by each firm participating in the auction, at the daily level before and after auction i results. Taking the day of auction i results as day 0, we calculate each auction participant's daily number of employees, starting from employment at day -84, up to day $+84$. Thus, the daily number of employees is firstly calculated to obtain 338 observations for each auction, 169 for the winner, and 169 for the runner-up, but to simplify the model, the mean number of employment is calculated for each fortnight, where fortnight is defined as *a period of two weeks.*[23] This yields 12 fortnights per firm-auction observation, where the base is the sixth fortnight preceding the auction results, zero (0) fortnight is the first two weeks after the auction, and the fifth fortnight is the mean employment growth between weeks 11 and 12 (compared with the base at sixth fortnight before the auction results). We take the log of the mean number of employees for each fortnight to take care of outliers.

The following equation is estimated:

$$Y_{ijt} = \alpha + \beta T_{ij} + \sum_{t=-5}^{+5} \gamma_t t_{ijt} + \sum_{t=-5}^{+5} \delta_{\mathrm{LATE}_t}(T_{ij} * t_{ijt}) + X_j^K \zeta_j + \epsilon_{ijt}, \tag{1}$$

where Y_{ijt} is the outcome variable (log employment) for a firm i at an auction j over the period t (i.e., from $t-6$ to $t+5$). T_{ij} is a dummy of one if the firm i won a PPC at the auction j, and zero if it is a runner-up. The t_{ijt} is a set of time dummies from -6 to 5, each representing a fortnight (a 14-day block) for a firm i calculated from the date of auction j. The X_j^K is a vector of control variables, including the firm-fixed effects, auction-fixed effects, lagged

place their bid just below the winning bid to maximize its project's profit. Thus, an estimate with potentially "*favored*" firms could be biased.

23 This approach is from Gugler et al. (2020) in which fortnight is defined as 2 *weeks* (also defined by the Cambridge Dictionary).

employment, political connection dummy, and winning margin, while ϵ_{ijt} is a random, normally distributed unobserved error term capturing factors that the model omits. The term δ is the *difference-in-differences* estimator, the average treatment effect on the treated of winning a PPC in the full sample of winners and runners-up, while it is the local average treatment effect for the subsample of close auctions. Standard errors are clustered at the firm level.

4. RESULTS

4.1 Public procurement auctions

The identification strategy focuses on competitive auctions with multiple bidders, but not all auctions have multiple bidders. Awarding PPCs via auctions with only a single bidder makes obtaining a valid counterfactual behavior for the analysis notoriously difficult. In general, submitting a bid on an auction signals that a firm is interested in undertaking the additional work and is growth-oriented.

A reflection on PPCs awarded via single-bidder auctions is provided in this subsection. Single bidding is considered a practice that should be avoided by the European Commission (Fazekas 2019), and some (notably Vuković 2020) have suggested single bidding auctions could represent the purchase of votes, insider information, corruption, and favorism.[24] However, single bidding practice was not uncommon in Croatia over the period 2016–18, when about 23.2%[25] of PPCs in construction were awarded based on single bidding auctions, while 76.8% are awarded based on auctions with two and more bidders. In value terms, total single bidder auctions were contracted at €938 million, and multiple bid auctions at €2792 million (descriptives in Tables A4–A6, and Figure A1). This study uses all auctions to show two descriptive analyses:

1) The relative cost savings from auction competition.
2) The value of single-bidder auctions marked with political connection variables.

Firstly, the benefits of auction competition for the contracting authority are shown. In particular, Figure 3 left panel depicts the ratio between the contracted price and the estimated PPC value across the number of bidders that applied for the construction works. Table A7 provides descriptive regression results, suggesting that in single-bidder auctions, the contracting authority pays for the contract, on average, 7.3% less than it estimates, while in cases in which there are eight bidders and more, the contracting authority pay on average 25.8% less than it estimates.[26] Figure 3 right panel focuses only on auctions with multiple bidders. *Money left on the table* (MLOTT, as in Gugler et al. 2020) is calculated, defined as the difference between the value of the runners-up bid (e.g., second-best option) and the value of the winning bid divided by the winning bid. Similar to the left panel, it shows that the average difference between the two best bids becomes smaller and smaller with the increase in the number of bids, providing

[24] Although considered bad practice, there can be a mix of justifiable reasons for having single-bidding auctions. In particular, procured works could be highly specialized and need specialized knowledge and equipment that only some firms are able to provide, while at the same time, there could be low supply and low interest for procured works, for reasons such as firms already working at their full working capacity, long geographical distance between the firm and the construction site, the excessively low estimated value of works, or other reasons.

[25] See Public Procurement Single Market Scoreboard, link: https://ec.europa.eu/internal_market/scoreboard/performance_per_policy_area/public_procurement/index_en.htm (accessed March 12, 2021). This considers auctions with one bid after the exclusion of invalid bids, in the entire dataset, we observe 15.33% of true single bid auctions, ones with no exclusions.

[26] Other factors might explain this ratio difference in auctions between single versus multiple bidders, including local economic conditions, high level of specialized knowledge, systematic errors in public authority estimation of the PPC value (denominator), or uniqueness of procured works. However, the descriptive regressions do not provide support for this reasoning and show similar cost savings as those in Figure 3 (Table A7).

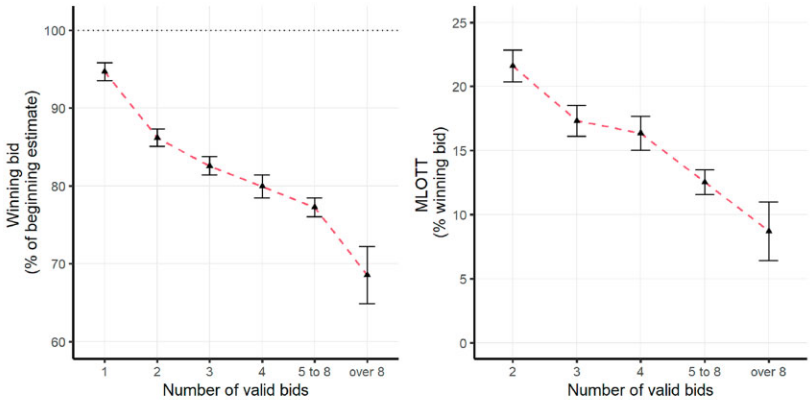

Figure 3. The relative costs of procured work by the number of bidders.

Notes: The left graph displays the winning bid as % of the auctions' estimated value (of both, LP and MEAT auctions). The graph shows the means and the lower and upper bounds of the confidence intervals (both on 95%). The right graph shows MLOTT as % of the auctions winning bid. Note that we analyze only valid bids. Both graphs exclude the top and bottom 1% of observations to account for outliers.

suggestive evidence confirming the theoretical model of Gugler et al. (2015), which suggests the bid is a function of the number of competitors, *inter alia* (Section 3). Thus, competition in auctions matters for the relative cost of construction works.

Secondly, single- and multiple-bidder auctions are used to show how much PPC value goes to firms marked as politically connected (Section 2.2.4 and Table A3). The study differentiates between the single- and multiple-bidder auctions and marks winning firms based on:

1) owners, CEOs, or supervisory board members who are politically connected individuals,[27]
2) donations to the political parties in power, and
3) single bid contract, which is more than 70% of the prior year's total revenue or if the winning firm has no employees.

The first marker identifies direct or indirect family ties of politicians with the firm owners. The general logic of how this marker is constructed at the local level is shown in Figure 2 while definitions are given in Table A3. The second marker provides a revealed preference perspective of firms' strategic networking choice or political stance. The third and final marker is described as *suspicious procurement* in the literature (as developed by Vuković 2020).[28] From the €490 million in single bid auctions in construction, our study shows that 82.7% of PPC value goes to either donators, politically connected firms, or suspicious firms (Table A4).[29] The same exercise is conducted on the sample of €2792 million awarded in

[27] It is elaborated how politically connected individuals are marked in Table A3, and Table A8 provides essential descriptives.

[28] It should be clearly stated that none of these indicators is evidence of corruption, but an indicator of *suspicious practice*. It should be noted that politically connected firms might be overstated in some cases. For example, in some municipalities or cities, there could be a mayor or council member with the same surname as owners, CEOs, or supervisory board members in a local construction firm, but they actually do not have family connections. To solve this, a correction for the 10 most frequent surnames in each municipality and region is conducted (as per Croatian Statistics Office: https://www.dzs.hr/hrv/censuses/census2011/results/htm/h01_01_31/H01_01_31.html [accessed August 17, 2021]).

[29] In total, 68% of value is marked to go to politically connected firms, 14.4% to firms donating to political parties, and 35.9% to suspicious winners (Table A4). The sum of these percentages goes above 100% because the firm could at the same time be a donator, a politically connected firm, and a suspicious firm.

multiple bid construction auctions and shows 65% of value goes to either donators, politically connected firms, or suspicious firms (Table A6). In multiple bid auctions, 37.4% (versus 68% in single bid) is marked to go to politically connected firms, 13.9% (versus 14.4% in single bid) goes to firms donating to political parties, and 32.5% (versus 35.9% in single bid) goes to suspicious firms. We then look at the auction level and code a dummy of one if the auction winner is a politically connected firm, and zero otherwise. We regress it on the number of bidders and also add controls for the baseline size of the winning firm and its other characteristics to see whether political connections are just picking up other firms' characteristics. We add the number of employees, year and month of auction result, and municipalities of the winner and buyer. Results in Table A9 show, compared with a single bidder, having four or more bidders in an auction is associated with 10 percentage points lower probability of a politically connected firm being the auction winner. In general, higher competition is associated with a lower probability of the auction winner being a politically connected firm.

In sum, we show auction competition is associated with both, lower relative costs of construction work and less frequently allocating contracts to politically connected firms. Next subsection questions whether auction competition can design a quasi-experimental setting to estimate the impact of PPC on firm employment.

4.2 Winners versus runners-up: balancing property

In the previous subsection, we show that competitive auctions are associated with lower relative costs and political influence. This section questions whether auction results can be considered a source of exogenous variation. Differences in the bids of auction participants can indicate differences in firms' production possibilities, political influence, or collusive behavior. This section provides a balancing property to shed light on this question, comparing winners, runners-up, and other auction participants on a long set of covariates.

We have 2859 auctions and 1071 unique firms over three years (2016–18).[30] Table 1 compares the means of all winners to all auction losers (ninth column), to auction participants with third, fourth, and fifth bids (seventh column), and runners-up (fifth column). Several insights occur. Compared with all auction losers (ninth column), winners are more frequently connected to a regional political party in power ($p < 0.001$) and national politicians ($p < 0.01$). Winners are geographically closer to the contracting authority ($p < 0.01$), have received more PPCs in the three years prior ($p < 0.05$), and are more frequently a state-owned enterprise (SOE; 7% versus 4%, $p < 0.001$). Relatedly, winners are slightly less productive ($p < 0.05$), have better financial indicators (EBITDA or profits over assets, $p < 0.01$), outsource less work, use less external labor (both at $p < 0.001$), and have more workers employed on a permanent work contract ($p < 0.001$). Differences decrease when the focus shifts only to winners and runners-up (fifth column), but winners are still geographically significantly closer to the public authority ($p < 0.001$) and are more frequently an SOE (7% versus 5%, $p < 0.05$).

Table 2 focuses on the subsample of close auctions (win margin within 10%) and compares the means of all winners to all auction losers (ninth column); to auction participants with third, fourth, and fifth bids (7th column); and to runners-up (fifth column). An improvement in balancing property is shown when the focus shifts only to winners and runners-up (fifth column versus seventh and ninth column). In particular, Table 2 shows no statistically significant difference in any of the analyzed covariates between the group of winners and runners-up (fifth column).

30 Auction and firm descriptives are available in Tables A10–A12.

Table 1. Comparison of winners, runner-ups, and others before the auction results.

Category (1)	Variable (2)	Winners (3)	Runners up (4)	Diff. (3)−(4) (5)	Ranks > 2 (6)	Diff. (3)−(6) (7)	Ranks > 1 (8)	Diff. (3)−(8) (9)
Employment	Observations	2859	2859		4514		7373	
	Employees	115.43	124.82	−9.39	125.19	−9.76	125.05	−9.62
	Employees, HE	18.23	19.22	−0.99	18.57	−0.34	18.82	−0.59
	Employees, LE	97.2	105.6	−8.4	106.62	−9.42	106.22	−9.03
	Hires	0.14	0.15	−0.01	0.16	−0.02	0.15	−0.01
	Fires	0.14	0.15	−0.01	0.15	−0.01	0.15	−0.01
	Tenure (in days)	1232.42	1246.32	−13.9	1169.29	63.13*	1199.14	33.28
	Employee age (in years)	35.15	35	0.15	35.18	−0.03	35.11	0.04
	Non-fixed term contracts (in %)	41.97	41.04	0.93	39.49	2.48***	40.09	1.88***
Projects	Backlog extensive	22.98	25.53	−2.55**	26.24	−3.26***	25.97	−2.98***
	Backlog intensive	12.67	13.39	−0.72	13.21	−0.54	13.28	−0.61
	Distance to contracting authority	68.45	77.17	−8.72**	72.28	−3.83	74.18	−5.73**
Political connections	Public firm	0.07	0.05	0.02**	0.04	0.03***	0.04	0.02***
	GONG/National match	0.17	0.15	0.01	0.15	0.02*	0.15	0.02*
	Regional match (in power)	0.24	0.22	0.02	0.2	0.04***	0.21	0.03**
	Local match (in power)	0.26	0.25	0.01	0.24	0.02	0.24	0.02
	Any match	0.47	0.47	0.01	0.46	0.01	0.46	0.01
	Any match (safe surnames)	0.41	0.4	0.01	0.39	0.02	0.39	0.01
	Any match (second order)	0.73	0.74	−0.01	0.73	0	0.74	0
	Donator	0.14	0.14	0	0.15	−0.01	0.15	−0.01
Balance sheet	Observations	2789	2757		4395		7152	
	Total assets	11.32	12.57	−1.24	11.98	−0.65	12.2	−0.88
	Current assets	5.73	5.85	−0.12	5.76	−0.03	5.79	−0.06
	Fixed assets	5.59	6.72	−1.13	6.22	−0.62	6.41	−0.82
	Total liabilities	5.17	6.1	−0.93	6.27	−1.1	6.2	−1.03
	Non-current liabilities	1.18	1.89	−0.7	1.88	−0.7	1.88	−0.7

(continued)

Table 1. (continued)

Category (1)	Variable (2)	Winners (3)	Runners up (4)	Diff. (3)−(4) (5)	Ranks > 2 (6)	Diff. (3)−(6) (7)	Ranks > 1 (8)	Diff. (3)−(8) (9)
Income statement	Revenue	12.17	12.29	–0.12	12.37	–0.21	12.34	–0.17
	EBITDA	1.17	1.09	0.08	0.96	0.21	1.01	0.16
	Profit	0.42	0.42	0	0.34	0.07	0.37	0.05
	Depreciation	0.51	0.43	0.08	0.36	0.15	0.39	0.12
	Interest paid	0.15	0.17	–0.02	0.18	–0.04	0.18	–0.03
	Wage costs	1.59	1.74	–0.15	1.7	–0.11	1.72	–0.12
	Productivity	0.11	0.12	–0.02	0.13	–0.02	0.12	–0.02*
Financial ratios	EBITDA over assets	0.13	0.12	0.01	0.12	0.01***	0.12	0.01**
	Profit over assets	0.06	0.05	0.01	0.05	0.01***	0.05	0.01*
	Debt ratio	0.55	0.63	–0.08	0.55	0	0.58	–0.03
	Outsourcing over total expenses	0.3	0.31	0	0.32	–0.02***	0.32	–0.01**
	External labor over total labor costs	0.33	0.33	–0.01	0.36	–0.04***	0.35	–0.03***

Notes: The first row represents the data after we exclude any auction in which the winner/runner-up is a firm for which we do not have the necessary employment data. The second notion of observations is a subset of those auctions, the one for which we have data on other financial data. For an explanation of all the variables, see Table A2. All monetary values are given in mil. Euro.
***, **, and * Significant at the 0.1%, 1%, and 5% level.

Table 2. Comparison of winners, runner-ups, and others before the auction results—in close auctions (within 10%).

Category (1)	Variable (2)	Winners (3)	Runners up (4)	Diff. (3)−(4) (5)	Ranks > 2 (6)	Diff. (3)−(6) (7)	Ranks > 1 (8)	Diff. (3)−(8) (9)
Employment	Observations	1436	1436		2719		4155	
	Employees	123.93	128.12	−4.19	123.28	0.65	124.95	−1.02
	Employees, HE	18.19	20.2	−2.01	18.84	−0.64	19.31	−1.12
	Employees, LE	105.74	107.91	−2.17	104.45	1.29	105.65	0.09
	Hires	0.15	0.15	0	0.16	0	0.16	0
	Fires	0.16	0.16	0	0.16	0	0.16	0
	Tenure (in days)	1279.07	1257.69	21.38	1203.43	75.64	1222.14	56.93
	Employee age (in years)	35.14	35.23	−0.09	35.33	−0.19	35.29	−0.15
	Non-fixed term contracts (in %)	39.15	40.07	−0.92	38.03	1.11	38.74	0.41
Projects	Backlog extensive	25.7	26.34	−0.64	27.75	−2.06	27.26	−1.57
	Backlog intensive	14.42	13.72	0.7	13.74	0.67	13.73	0.68
	Distance to contracting authority	69.58	73.65	−4.07	72.03	−2.46	72.59	−3.01
Political connections	Public firm	0.07	0.06	0.01	0.04	0.02**	0.05	0.02**
	GONG/National match	0.19	0.16	0.03	0.15	0.04**	0.16	0.03**
	Regional match (in power)	0.24	0.22	0.02	0.21	0.03*	0.21	0.03*
	Local match (in power)	0.26	0.25	0.01	0.26	0.01	0.26	0.01
	Any match	0.49	0.47	0.02	0.47	0.02	0.47	0.02
	Any match (safe surnames)	0.43	0.4	0.03	0.4	0.02	0.4	0.02
	Any match (second order)	0.75	0.73	0.02	0.74	0.01	0.74	0.01
	Donator	0.16	0.15	0.01	0.15	0.01	0.15	0.01
Balance sheet	Observations	1399	1370		2639		4009	
	Total assets	11.87	14.18	−2.31	11.43	0.44	12.37	−0.5
	Current assets	5.82	6.32	−0.49	5.88	−0.06	6.03	−0.21
	Fixed assets	6.05	7.86	−1.81	5.55	0.5	6.34	−0.29
	Total liabilities	5.87	6.71	−0.84	6.22	−0.35	6.39	−0.52
	Non-current liabilities	1.27	2.21	−0.94	1.73	−0.46	1.9	−0.62

(continued)

Table 2. (continued)

Category (1)	Variable (2)	Winners (3)	Runners up (4)	Diff. (3)−(4) (5)	Ranks > 2 (6)	Diff. (3)−(6) (7)	Ranks > 1 (8)	Diff. (3)−(8) (9)
Income statement	Revenue	13.19	12.49	0.7	12.6	0.58	12.56	0.62
	EBITDA	1.11	1.22	–0.11	0.92	0.18	1.02	0.08
	Profit	0.34	0.44	–0.1	0.31	0.03	0.36	–0.01
	Depreciation	0.51	0.5	0.01	0.35	0.15	0.4	0.11
	Interest paid	0.17	0.2	–0.03	0.19	–0.02	0.19	–0.02
	Wage costs	1.69	1.76	–0.07	1.64	0.05	1.68	0.01
	Productivity	0.11	0.13	–0.01	0.13	–0.01	0.13	–0.01
Financial ratios	EBITDA over assets	0.13	0.11	0.02	0.12	0.01***	0.11	0.01
	Profit over assets	0.05	0.04	0.01	0.05	0.01*	0.04	0.01
	Debt ratio	0.55	0.57	–0.02	0.55	0	0.56	–0.01
	Outsourcing over total expenses	0.31	0.3	0	0.32	–0.02**	0.31	–0.01
	External labor over total labor costs	0.33	0.32	0.01	0.36	–0.03**	0.35	–0.02

Notes: The first row represents the data after we exclude any auction in which the winner/runner-up is a firm for which we do not have the necessary employment data. The second notion of observations is a subset of those auctions, the one for which we have data on other financial data. For an explanation of all the variables, see Table A2. All monetary values are given in mil. Euro.
***, **, and * Significant at the 0.1%, 1%, and 5% level.

Achieving balancing property does not mean winners are not politically connected but are, on average, similarly politically connected as runners-up. About 24% of the winning firms are connected to regional/county politicians in power (compared with 22% of the runners-up), and 26% to local politicians in power (compared with 25% of the runners-up). Winners and runners-up groups are, on average, donators to the political party in similar share (16% versus 15%) and are similarly connected to national politicians (19% versus 16%). Winners and runners-up groups are also, on average, similar in size (124 versus 128 employees), have, on average, about 20 employees with tertiary education, where the average employee tenure is 3.5 years, and the average employee age at the day of employment is 35 years. Both groups have a debt ratio of about 55%, and about 31% of total expenses go to outsourcing, while external workers are about 33% of total labor costs (for details, see Table 2, third, fourth, and fifth columns).

4.3 Information leakage channel

Winners and runners-up have, on average, similar geographical distance to the public procurement authority (i.e., buyer) and have similar political connections. Previous research (i.e., Baltrunaite 2020; P. Andreyanov, A. Davidson, and V. Korovkin, unpublished data) suggests an information leakage channel by which contracting authorities provide confidential information to favored firms winning the PPCs. If this was so, there could be a selection of less productive, politically connected firms into PPCs, potentially charging a higher price for construction work—leading to a higher cost per job.[31] We analyze the information leakage issue by collecting data on the timing of each bid. If information leakage holds, there could be systematic sorting of winners, especially winners with political connections who place bids after runners-up. We visually show the difference in the timing of the bids between winners and runners-up (Figures A2 and A3), while in Table A13 we show data for the difference in the timing of the bids. We code a dummy of 1 if the winner places the bid after the runner-up and vice versa. Results indicate that there is no systematic sorting of winners into placing bids after runners-up, which is the same regardless of the characteristics of the auction (i.e., larger contracts) or subgroup of firms (i.e., politically connected firms). The finding of no sorting is different from the analysis in Russia (i.e., P. Andreyanov, A. Davidson, and V. Korovkin, unpublished data). However, during their analysis period in Russia, there was no encryption of bids in the auction system. In sum, we find no compelling evidence to explain the information leakage channel as an explanation for politically connected winning auctions (Table A14).[32]

4.4 The impact of PPC on firms' employment

Upon evaluating the balancing property and information leakage channel, we visually inspect the employment growth trends of winners and runners-up (Figure A4). Winning firms increase their employment after receiving a PPC. However, runners-up also experience employment growth, but at a lower rate than winners. The general upward-moving trend in employment among the two groups is probably due to the expansion period characterized by a significant decrease in unemployment and a sharp increase in construction demand and government consumption (see also Figure 1). The visual trends in Figure A4 are quantified with Equation (1) and show a positive impact of PPC on firms' employment (Table 3).

[31] This is quite difficult to document in construction, since every project is, in a sense, unique (because there are no repeating locations).

[32] We also conducted interviews on the information leakage channel with two experts, one in law, and one information technology expert on PPC auctions (see Appendix B).

Table 3. Main regression estimates.

	All (1)	Within 10% (main) (2)	Within 8% (3)	Within 6% (4)	Within 4% (5)	4–0.5% (6)
−5	–0.003	0.013	0.067	–0.076	0.042	0.203
	(0.544)	(0.572)	(0.593)	(0.664)	(0.775)	(0.860)
−4	0.073	0.064	0.055	–0.030	0.153	0.391
	(0.544)	(0.572)	(0.593)	(0.664)	(0.775)	(0.860)
−3	0.182	0.106	0.047	–0.082	0.123	0.258
	(0.544)	(0.572)	(0.593)	(0.664)	(0.775)	(0.860)
−2	0.400	0.453	0.403	0.161	0.430	0.302
	(0.544)	(0.572)	(0.593)	(0.664)	(0.775)	(0.860)
−1	0.500	0.440	0.303	0.042	0.380	0.358
	(0.544)	(0.572)	(0.593)	(0.664)	(0.775)	(0.860)
0	0.995*	1.012*	0.834	0.471	0.808	0.846
	(0.544)	(0.572)	(0.593)	(0.664)	(0.775)	(0.860)
1	1.386***	1.454***	1.408**	0.898	0.885	0.962
	(0.544)	(0.572)	(0.593)	(0.664)	(0.775)	(0.860)
2	1.795***	1.661***	1.689***	1.308**	1.234	1.345
	(0.544)	(0.572)	(0.593)	(0.664)	(0.775)	(0.860)
3	2.471***	2.333***	2.301***	2.110***	2.129***	2.069**
	(0.544)	(0.572)	(0.593)	(0.664)	(0.775)	(0.860)
4	2.546***	2.569***	2.378***	2.171***	2.538***	2.602***
	(0.544)	(0.572)	(0.593)	(0.664)	(0.775)	(0.860)
5	2.521***	2.703***	2.474***	2.175***	2.630***	2.998***
	(0.544)	(0.572)	(0.593)	(0.664)	(0.775)	(0.860)
Mean employees	119.4948	124.2909	120.9615	122.5415	123.4315	122.0516
N	62,544	31,872	27,720	21,912	16,140	13,332
R^2	0.516	0.629	0.631	0.647	0.651	0.654
Adjusted R^2	0.414	0.544	0.544	0.561	0.564	0.565
Residual Std. Error	10.143 (df = 51,659)	7.563 (df = 25,934)	7.296 (df = 22,458)	7.254 (df = 17,648)	7.264 (df = 12,904)	7.330 (df = 10,609)

Notes: Column (1) shows the estimates for the full sample. Other columns—subsamples of close auctions are constructed according to the winning margin. The win margin of 10%, 8%, 6%, 4%, and 4–0.5% is based on the win margin definition unless stated otherwise (see "Methods). The dependent variable is employment growth at firm–auction level in each fortnight period (from −6 to 5). The model is estimated with Equation (1). The independent variables are the fortnight periods, win margin, auction- and firm-fixed effects, political connection dummy, as well as the (log) number of firms' employees in the −sixth fortnight. The estimates are calculated using the package ("lfe," Gaure 2013) and show the LATE, difference in employment growth rates between winners and runner-ups. The point estimates and standard errors are transformed to absolute employment increases based on the coefficients and the mean number of employees (given in "mean employees") in the −sixth fortnight. ***, **, and * Significant at the 1%, 5%, and 10% level.

The full sample of winners and runners-up shows a positive effect of 2.5 employees (±0.5) in the fifth fortnight (70–84 days post-PPC; Table 3, first column), while the main regression estimates (Table 3, second column) show a somewhat higher positive effect of 2.7 employees (±0.6) at the fifth fortnight. The difference between the full sample and close auctions could be in lower transportation costs and potentially better information, as well as more state-owned firms benefiting from the contracts (i.e., Table 1). Winning firms in the full sample might have more organizational slack and less need to hire new employees. Runners-up might grow more from the private market upon losing the contract. The results for the close auction sample are similar regardless of whether close auctions are defined by the winning margin within 8%, 6%, 4%, or 0.5–4% (third, fourth, fifth, and sixth columns, Table 3).[33] We gradually add control variables in a close auction sample, from a simple difference-in-differences model to our main model, including firm-fixed effects, auction-fixed effects, lagged employment, winning margin, and political connections dummy. Table A15 shows that results are stable when adding control variables.

In the main model, the effect starts to be weakly significant in 14–28 days after the auction results (fortnight 1), and becomes larger in magnitude from the second fortnight onwards, while the point estimate is highest and highly significant ($p < 0.01$) in the fifth fortnight (70–84 days) after the winning notice in all models (panels b–f, Figure 4).

In Section 5, the magnitude of estimates and public cost per job directly created are discussed. The next section analyses the robustness of the main model.

4.5 Robustness

The main model is based on several assumptions regarding competitive auctions, supplier selection methods, and the intention of runners-up to work on more projects. In the rest of this subsection, several robustness analyses are conducted to test the stability of the main model estimates.

4.5. 1 Auction complaints

If auction participants suspect there might be favoritism in connection with the technical documentation of procured work, they can submit a complaint to DKOM. Table A16 shows frequent complaints by the procurements' first four-digit CPV code and the top 10 contracting authorities that received the most complaints. About the complaints by the procurements' CPV, the code starting with 4523 (which represents procurements related to the construction of highways and roads) received 32% of all construction complaints.[34] For a robustness check, two regressions are run. First, regression without auctions of works for four-digit CPVs with the most frequent complaints shows a statistically significant and positive impact of the PPC on firms' employment with a somewhat higher magnitude of 5.2 employees (±1.3). Second, regression without firms located in the City of Zagreb shows the result remains significant and of similar magnitude to the main model estimate (Table 4, Columns 2 and 3).

4.5. 2 Donations to political parties

Donations to political parties have been identified as important in acquiring PPCs (Baltrunaite 2020). This study examines donations to political parties preceding the parliamentary elections

[33] After only excluding the auctions for which we do not have the employment data on winners or runner-ups there is a total of 2859 auctions (10,232 bids). The close auction 10% win margin then has 1436 auctions (5591 bids), 8% win margin has 1250 auctions (4897 bids), 6% win margin has 994 auctions (3927 bids), 4% win margin has 737 auctions (2932 bids), and 0.5–4% win margin has 610 auctions (2470 bids).

[34] The highest number of complaints are registered for the City of Zagreb (4%), Hrvatske ceste d.o.o. (3%), and other SOE with headquarters in Zagreb (see Figure A5 and Table A16).

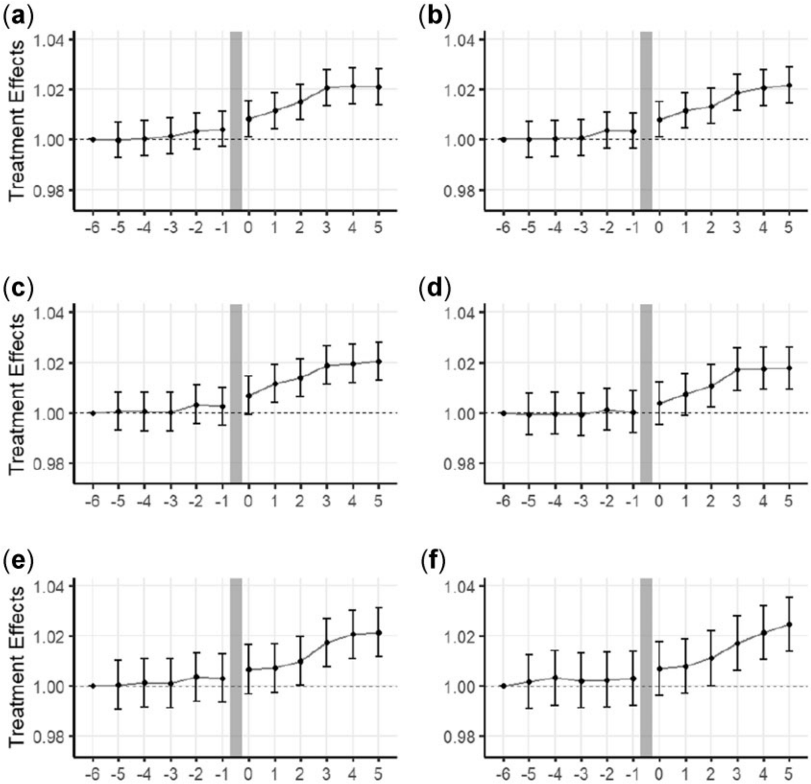

Figure 4. The impact of PPC on firm employment.

Notes: The point estimates with confidence intervals at 90% are extracted from the main model (Table 3). The panels (**a**) includes all winners and runners-up, (**b**) includes auctions where the difference between the winners and runners-up is up to 10%, (**c**) auctions up to 8% difference, (**d**) auctions up to 6% difference, (**e**) auctions up to 4% difference, (**f**) auctions with a difference from 0.5% to 4% between winners and runners-up.

in 2015 and 2016. These donators to the political parties placed 1507 of 10,232 bids (descriptives in Table A17). A model without firms donating to the political parties is run, and importantly, shows that the main model estimates do not change (Table 4, Column 6).

4.5. 3 Politically connected firms

With the political connection proxy (see Table A3), we mark 373 politically connected firms. They place 1346 winning bids in auctions. A model accounting for all the politically connected firms is run and shows the main model estimates remain robust (Table 4, Column 4). A potential limitation of the local- and regional-level political connection proxies is its reliance on matching politicians to firms with owners, CEOs, and/or supervisory board members holding the same surname within the geographical (municipal/regional) location. To check whether results are driven by politically connected proxy being overly wide, all the politically connected firms that have owners, CEOs, or supervisory board members whose surnames are on the list of most frequent surnames in the respective municipality or region are recorded as not politically connected.[35] All the recorded politically connected firms are

[35] Based on Croatian Statistics Office: https://www.dzs.hr/hrv/censuses/census2011/results/htm/h01_01_31/H01_01_31.html (accessed August 17, 2021).

Table 4. Robustness checks.

	Main (within 10%) (1)	No CPVs with most comp. (2)	No region with most comp. (3)	No pol. connections (4)	No pol. conn. (robust surnames) (5)	No pol. donators (6)	No suspicious firms (7)
−5	0.013	0.629	0.003	0.010	0.010	0.082	0.065
	(0.572)	(1.268)	(0.702)	(0.572)	(0.572)	(0.577)	(0.577)
−4	0.064	0.506	0.248	0.032	0.032	0.312	0.153
	(0.572)	(1.268)	(0.702)	(0.572)	(0.572)	(0.577)	(0.577)
−3	0.106	0.792	0.194	0.030	0.030	0.364	0.130
	(0.572)	(1.268)	(0.702)	(0.572)	(0.572)	(0.577)	(0.577)
−2	0.453	1.114	0.660	0.358	0.358	0.679	0.439
	(0.572)	(1.268)	(0.702)	(0.572)	(0.572)	(0.577)	(0.577)
−1	0.440	1.485	0.491	0.349	0.349	0.745	0.369
	(0.572)	(1.268)	(0.702)	(0.572)	(0.572)	(0.577)	(0.577)
0	1.012*	2.193*	1.156*	0.924*	0.924*	1.341**	0.766
	(0.572)	(1.268)	(0.702)	(0.572)	(0.572)	(0.577)	(0.577)
1	1.454***	2.504**	1.845***	1.373**	1.373**	1.799***	0.994*
	(0.572)	(1.268)	(0.702)	(0.572)	(0.572)	(0.577)	(0.577)
2	1.661***	3.114**	2.060***	1.576***	1.576***	2.016***	1.140**
	(0.572)	(1.268)	(0.702)	(0.572)	(0.572)	(0.577)	(0.577)
3	2.333***	4.395***	2.554***	2.264***	2.264***	2.583***	1.876***
	(0.572)	(1.268)	(0.702)	(0.572)	(0.572)	(0.577)	(0.577)
4	2.569***	4.404***	2.877***	2.537***	2.537***	2.832***	1.991***
	(0.572)	(1.268)	(0.702)	(0.572)	(0.572)	(0.577)	(0.577)
5	2.703***	5.227***	2.784***	2.677***	2.677***	2.980***	1.924***
	(0.572)	(1.268)	(0.702)	(0.572)	(0.572)	(0.577)	(0.577)
Mean emp.	124.2909	116.1932	121.666	124.2909	124.2909	124.2909	124.2909
N	31,872	7776	22,800	31,872	31,872	31,872	31,872
R^2	0.629	0.615	0.630	0.629	0.629	0.629	0.628
Adj. R^2	0.544	0.514	0.542	0.544	0.543	0.544	0.543
Residual Std. Error	7.563 (df = 25,934)	8.651 (df = 6150)	7.381 (df = 18,409)	7.564 (df = 25,945)	7.565 (df = 25,934)	7.561 (df = 25,934)	7.569 (df = 25,934)

Notes: Column (1) shows the estimates for the main sample. Columns (2) and (3) show estimates without frequent complaints. The top five CPV 4-digit codes with the most complaints are 4523, 4521, 4545, 4500, and 4524, which are excluded from the regression in Column (2), and the county with the most complaints is the City of Zagreb, hence we exclude all auctions where a winner is headquartered in it and show the estimates in Column (3). Column (4) uses the main sample with 10%, but the win dummy is only 1 if the winner is not a politically connected firm. Column (5) does the same but uses a "surname"-robust political connection dummy. Column (6) does the same but with a dummy if a firm donated to any political party. Column (7) repeats but with a dummy for a suspicious firm (see Table A4). The dependent variable is employment growth at firm–auction level in each fortnight period (from −6 to 5). The model is estimated with Equation (1). The independent variables are the fortnight periods, win margin, auction- and firm-fixed effects, political connection dummy, as well as the (log) number of firms' employees in the −sixth fortnight. The estimates are calculated using the package ("lfe," Gaure 2013) and show the LATE, difference in employment growth rates between winners and runner-ups. The point estimates and standard errors are transformed to absolute employment increases based on the coefficients and the mean number of employees (given in "mean employees") in the −sixth fortnight.
***, **, and * Significant at the 1%, 5%, and 10% level.

accounted for and regression is run to show similar effects as in the main model (Table 4, Column 5).

4.5. 4 LP and MEAT

The analysis encompasses the years 2016–18. From January 1, 2016 to July 1, 2017, the auctions were governed by the Public Procurement Law (NN 90/2011), while from July 1, 2017 onward it has been governed by the new Public Procurement Law (NN 120/16). The new law (NN 120/16) introduced mandatory usage of the MEAT, which in essence required the selection of suppliers to be not only based on the LP criteria but the criteria related to suppliers' quality too. To do so, the law capped the maximum share of points a supplier could achieve based on price to 90%.

The descriptive analysis reported in Tables A18 and A19 shows about 61% of auctions are awarded based on the LP and 39% based on the MEAT criteria. Auctions governed by the law (NN 90/2011) and those governed by the *new* law (NN 120/16) have a similar mean PPC size (€730,000 versus 710,000, w/o VAT). After the new law, contracting authorities did combine price criteria with quality criteria, usually operationalized as *quality*, *costs*, and/or *time* of finalizing the construction work. However, two findings emerge. First, time to complete construction is the most commonly used quality criteria.[36] Second, while the law introduced a mandatory shift to MEAT, 61% of auctions under the new law use the maximum 90% weight on price criteria, 28% of auctions use price criteria weight between 80% and 89%, 7% of auctions use price criteria weight between 70% and 79%, and only 4% with price criteria weight below 70%. Results in Table 5 show a positive impact of PPC on firms' employment regardless of whether LP (price) or MEAT criteria (points) are used to calculate the winning margin.

4.5. 5 Contamination of evaluation period and seasonality

Contamination of the evaluation period can be important, as winning firms might win multiple PPCs over the evaluation period, and could win more PPCs than the runners-up during the evaluation period (or vice versa). A regression controlling for the number of contracts won by the winner and by the runner-up during the evaluation periods is run, and the results do not change (Table A20). In addition, a model is run on the subsample of close auctions where the winner receives only a single PPC during the evaluation period and runner-ups receive no PPC. Table A20 shows that the results do not change.

Finally, the impact of PPC on firms' employment might be driven by seasonality: on the one hand, employment could increase systematically more in those months when the weather is warmer and less rainy, while on the other hand, employment could decline or stagnate in winter months when construction works are less intensive (Gugler et al. 2020). Seasonality could lead to correlations of seasonal employment change with treatment status. The sample is split to construct in-season and off-season subsamples; the first encompasses April–October, while the second includes November–March. Results in Table A20 (Columns 2 and 3) show the effects are of similar magnitude in two different seasons. In addition, a regression with an added dummy for the month of the auction results shows results similar to the main model estimates (Table A20).

[36] Which is a relatively easier indicator to obtain points for the majority of construction firms?

Table 5. The impact of PPC on firms' employment: LP and MEAT samples.

		LP sample		MEAT sample	
	Entire sample	All	Within 10%	All	Within 10%
	(1)	(2)	(3)	(4)	(5)
−5	−0.003	0.026	0.213	−0.047	0.151
	(0.544)	(0.512)	(0.612)	(1.139)	(1.163)
−4	0.071	0.043	0.248	0.114	0.356
	(0.544)	(0.512)	(0.612)	(1.139)	(1.163)
−3	0.173	−0.007	0.290	0.463	0.777
	(0.544)	(0.512)	(0.612)	(1.139)	(1.163)
−2	0.388	0.615	0.843	0.076	0.239
	(0.544)	(0.512)	(0.612)	(1.139)	(1.163)
−1	0.496	0.913*	0.838	−0.107	−0.171
	(0.544)	(0.512)	(0.612)	(1.139)	(1.163)
0	0.994*	1.446***	1.433**	0.329	−0.007
	(0.544)	(0.512)	(0.612)	(1.139)	(1.163)
1	1.385***	1.798***	1.728***	0.787	0.593
	(0.544)	(0.512)	(0.612)	(1.139)	(1.163)
2	1.792***	2.046***	1.756***	1.422	1.213
	(0.544)	(0.512)	(0.612)	(1.139)	(1.163)
3	2.475***	2.439***	2.266***	2.491**	2.938***
	(0.544)	(0.512)	(0.612)	(1.139)	(1.163)
4	2.553***	2.369***	2.393***	2.747**	3.244***
	(0.544)	(0.512)	(0.612)	(1.139)	(1.163)
5	2.530***	2.263***	2.469***	2.811**	3.551***
	(0.544)	(0.512)	(0.612)	(1.139)	(1.163)
Mean employees	119.4948	122.5222	126.6743	114.5732	117.7196
N	62,544	38,724	27,228	23,820	15,816
R^2	0.516	0.628	0.633	0.437	0.501
Adjusted R^2	0.414	0.546	0.548	0.301	0.375
Residual Std. Error	10.143 (df = 51,659)	7.507 (df = 31,688)	7.481 (df = 22,105)	13.143 (df = 19,208)	10.858 (df = 12,638)

Notes: The margin for bid closeness is calculated with price in the LP sample and points in the MEAT sample. The dependent variable is employment growth at firm–auction level in each fortnight period (from −6 to 5). The model is estimated with Equation (1). The independent variables are the fortnight periods, auction-, and firm-fixed effects, as well as the number of firms' employees in the −sixth fortnight, which are included. The estimates are calculated using the package ("lfe," Gaure 2013) and show the LATE, difference in employment growth rates between winners and runner-ups in the close auction sample. The point estimates and standard errors are transformed to absolute employment increase based on the coefficients and the mean number of employees (given in "mean employees") in the −sixth fortnight.
***, **, and * Significant at the 1%, 5%, and 10% level.

5. MECHANISMS AND HETEROGENEITY

The main model (Section 4.4) estimates a positive impact of PPC on firms' employment in the magnitude of 2.7 employees (±0.6). On a sample of small firms in retail and manufacturing in Brazil, Ferraz et al. (2021) find the effect of 0.63 employees (about 2.2% employment growth), while on a sample of larger construction firms in Austria, Gugler et al. (2020) show growth of about 3% of the winners' workforce or about 80 employees. To better understand the magnitude, median PPC size (with VAT) is divided by the absolute effect size to obtain the initial public cost per job directly created at €93,223, which is considerably higher than the point estimates in similar studies in Austria (€22,246 in Gugler et al. 2020) and Brazil ($58,807 in Ferraz et al. 2021), but smaller than PPC estimates with more aggregate data in the United States ($125,000 in Wilson 2012). In a related field on the impact evaluations of public grants for firms, a recent review in the European Union shows public cost per job directly created to a range from €189,000, €62,000, €26,000, €14,700 to €6000 (Dvouletý et al. 2021: 256). In Croatia, for self-employment grants, Srhoj and Zilic (2021) estimate €38,800 public cost per job, and for small- and medium-sized enterprise grants, Srhoj et al. (2021) estimate €14,525 public cost per job. Therefore, in this subsection, analyses are focused on the heterogeneity of effects, and mechanisms that can explain the initial high estimate of the cost per job (€93,223).

5.1 Worker characteristics

Descriptive characteristics of the winners' 11,285 new (unique) employees in the fifth fortnight are provided in Table A21. A lion's share of new employees works in positions requiring low education (89%). The five most common professions among new employees are masons (9%), carpenters and joiners (7%), workers without occupations (7%), civil engineering workers (6%), and operators of construction and similar machines (6%) (Table A22). Of the 11,285 new employees, 46% worked for the winning firm during their last employment episode, 14% worked for a different firm in their last employment episode, while a substantial 40% are registered to HZMO for the first time. From the 14% of employees that worked for another firm, Table A23 shows the majority come from the construction sector (73%), manufacturing (9%), wholesale and retail trade (7%), and all other sectors (11%). The 40% registered at HZMO for the first time can happen for two reasons: (i) domestic individuals' first employment or (ii) foreign individuals' first employment in Croatia, which does not mean they had not worked before in their country of origin. The dataset does not enable us to identify foreign nationals but since the group of employees with first registered employment has 34.4 years on average, there is probably a substantial share of employees who are not first-time domestic hires, but more probably foreign nationals employed for construction projects. Thus, descriptively, new hires among winning firms are workers previously working in the construction sector as bricklayers, workers without occupations, carpenters, and joiners, with about 40% of winners' new employees having their first employment in the country from which a non-negotiable share are probably foreign workers.

We begin the causal heterogeneity analysis by examining worker characteristics. In particular, we examine whether firms employ predominantly male or female workers, on fixed or non-fixed-term contracts, and jobs requiring higher or lower education. Results in Table 6 show the majority of positive effect stems from employing male workers, on fixed-term work contract, and on job positions requiring lower education. Next, we examine heterogeneous effects with respect to employment history. We examine whether new employees are dominantly (i) hired for the first time in the county, (ii) hired after a longer unemployment (longer than 30 days), and (iii) without unemployment or after a short unemployment spell (less than 30 days). Table 7 shows the highest PPC positive impact for employing workers who have their first work contract in the country. There is also a positive impact on employing workers who are already working (e.g., in the same, or another firm) or after a short unemployment spell (up to 30 days). The lowest magnitude of the impact is documented for the workers who had a longer unemployment spell prior to employment.

In sum, the anatomy of identified new hires shows the majority are male, hired on fixed-term contracts, and in job positions requiring lower education. Additionally, the majority of new hires are working in the country for the first time, from which a non-negotiable share is probably foreign workers. Among those new hires that are not employed for the first time, the majority are transferred from other construction firms or workers experiencing a brief unemployment spell.

Table 6. Effects on employment of different types of employees.

		Needed education level		Employee sex		Employee contract	
	Main	Higher	Lower	Female	Male	Non-fixed term	Fixed term
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
−5	0.013	−0.024	−0.109	0.067	−0.182	−0.098	0.003
	(0.572)	(0.113)	(0.566)	(0.083)	(0.535)	(0.269)	(0.868)
−4	0.064	0.035	−0.118	0.036	−0.042	−0.115	0.157
	(0.572)	(0.113)	(0.566)	(0.083)	(0.535)	(0.269)	(0.868)
−3	0.106	0.065	−0.195	−0.001	0.149	−0.095	0.458
	(0.572)	(0.113)	(0.566)	(0.083)	(0.535)	(0.269)	(0.868)
−2	0.453	0.119	0.006	0.059	0.503	−0.104	0.861
	(0.572)	(0.113)	(0.566)	(0.083)	(0.535)	(0.269)	(0.868)
−1	0.440	0.157	−0.049	0.127	0.431	−0.009	0.940
	(0.572)	(0.113)	(0.566)	(0.083)	(0.535)	(0.269)	(0.868)
0	1.012*	0.143	0.511	0.104	0.984*	0.081	1.903**
	(0.572)	(0.113)	(0.566)	(0.083)	(0.535)	(0.269)	(0.868)
1	1.454***	0.140	0.877	0.102	1.423***	0.226	2.896***
	(0.572)	(0.113)	(0.566)	(0.083)	(0.535)	(0.269)	(0.868)
2	1.661***	0.136	0.891	0.074	1.699***	0.230	3.956***
	(0.572)	(0.113)	(0.566)	(0.083)	(0.535)	(0.269)	(0.868)
3	2.333***	0.158	1.386**	0.067	2.375***	0.531**	4.855***
	(0.572)	(0.113)	(0.566)	(0.083)	(0.535)	(0.269)	(0.868)
4	2.569***	0.175	1.600***	0.118	2.540***	0.720***	4.874***
	(0.572)	(0.113)	(0.566)	(0.083)	(0.535)	(0.269)	(0.868)
5	2.703***	0.228**	1.639***	0.231***	2.447***	0.951***	4.607***
	(0.572)	(0.113)	(0.566)	(0.083)	(0.535)	(0.269)	(0.868)
Mean employees	124.2909	19.0894	105.1597	14.1782	110.0374	40.0958	84.1895
N	31,872	31,872	31,872	31,872	31,872	31,872	31,872
R^2	0.629	0.635	0.625	0.597	0.634	0.673	0.646
Adj. R^2	0.544	0.551	0.539	0.505	0.550	0.598	0.565
Residual Std. Error (df = 25,934)	7.563	1.493	7.482	1.098	7.072	3.553	11.480

Notes: Requirements for higher education level job include any of the following: specialized doctorate, doctorate, a master's degree (and specialized masters programs), bachelor's degree. The dependent variable is the employment growth for each specific worker type given for each column, at a firm–auction level in each fortnight (from −6 to 5). The model is estimated with Equation (1). The independent variables are the fortnight periods, win margin, auction- and firm-fixed effects, political connection dummy, as well as the (log) number of firms' employees in the −sixth fortnight. The estimates are calculated using the package ("lfe," Gaure 2013) and show the LATE, difference in employment growth rates between winners and runner-ups. The point estimates and standard errors are transformed to absolute employment increases based on the coefficients and the mean number of employees (given in "mean employees") in the −sixth fortnight.
***, **, and * Significant at the 1%, 5%, and 10% level.

5.2 Longer term effects

The effects on employment are small in magnitude, winning firms hire low-educated male workers, on fixed-term contracts, with probably a substantial share being foreign nationals. However, the effects could be persistent over time, for example, because winners take fewer workers but over a longer period to complete the work. We estimate a similar regression to Equation (1):

$$Y_{ijt} = \alpha + \beta T_{ij} + \sum_{t=-5}^{+20} \gamma_t t_{ijt} + \sum_{t=-5}^{+20} \delta_{\mathrm{LATE}_t} (T_{ij} * t_{ijt}) + X_j^K \zeta_j + \epsilon_{ijt}, \tag{2}$$

Table 7. Effects on employment of different types of employees.

		Across all sectors			Only in the construction sector		
	Main	First time employment	Employment after more than 30 days of being unemployed	Employment within 30 days from last employment	First time employment	Employment after more than 30 days of being unemployed	Employment within 30 days from last employment
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
−5	0.013	−0.185	−0.002	0.086	−0.088	−0.027	−0.120
	(0.572)	(0.440)	(0.247)	(0.394)	(0.554)	(0.174)	(0.330)
−4	0.064	−0.224	−0.043	0.204	−0.301	−0.060	0.007
	(0.572)	(0.440)	(0.247)	(0.394)	(0.554)	(0.174)	(0.330)
−3	0.106	−0.409	0.009	0.367	−0.719	−0.011	0.102
	(0.572)	(0.440)	(0.247)	(0.394)	(0.554)	(0.174)	(0.330)
−2	0.453	−0.351	0.095	0.449	−0.543	0.057	0.107
	(0.572)	(0.440)	(0.247)	(0.394)	(0.554)	(0.174)	(0.330)
−1	0.440	−0.115	0.125	0.195	−0.496	0.002	−0.010
	(0.572)	(0.440)	(0.247)	(0.394)	(0.554)	(0.174)	(0.330)
0	1.012*	0.557	0.179	0.144	0.096	0.073	−0.007
	(0.572)	(0.440)	(0.247)	(0.394)	(0.554)	(0.174)	(0.330)
1	1.454**	1.021**	0.381	0.210	0.719	0.091	0.244
	(0.572)	(0.440)	(0.247)	(0.394)	(0.554)	(0.174)	(0.330)
2	1.661***	1.166***	0.415*	0.345	0.947*	0.091	0.373
	(0.572)	(0.440)	(0.247)	(0.394)	(0.554)	(0.174)	(0.330)

(continued)

Table 7. (continued)

		Across all sectors			Only in the construction sector		
	Main	**First time employment**	**Employment after more than 30 days of being unemployed**	**Employment within 30 days from last employment**	**First time employment**	**Employment after more than 30 days of being unemployed**	**Employment within 30 days from last employment**
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
3	2.333***	1.515***	0.549**	0.725*	1.461***	0.168	0.651**
	(0.572)	(0.440)	(0.247)	(0.394)	(0.554)	(0.174)	(0.330)
4	2.569***	1.774***	0.587**	0.786**	1.758***	0.154	0.697**
	(0.572)	(0.440)	(0.247)	(0.394)	(0.554)	(0.174)	(0.330)
5	2.703***	1.653***	0.509**	0.954**	1.768***	0.127	0.728**
	(0.572)	(0.440)	(0.247)	(0.394)	(0.554)	(0.174)	(0.330)
Mean employees	124.2909	78.4955	11.2528	34.5295	88.7908	8.3511	27.1467
N	31,872	31,872	31,872	31,872	31,872	31,872	31,872
R^2	0.629	0.596	0.667	0.661	0.652	0.685	0.669
Adjusted R^2	0.544	0.504	0.591	0.583	0.572	0.613	0.593
Residual Std. Error (df = 25,934)	7.563	5.821	3.272	5.219	7.324	2.296	4.368

Notes: Columns (2)–(4) take into account all available data on employment. Columns (5)–(7) only look at the construction sector, which is all firms in our sample plus all other firms which are classified as "F" according to NACEREV-21. As such, Column (2) only looks at employees which appear for the first time in the HZMO database, while Column (5) looks at only the employees which were not employed in any construction sector firms previously. The same principle holds in other columns. The dependent variable is the employment growth for each specific worker type given for each column, at a firm–auction level in each fortnight (from −6 to 5). The model is estimated with Equation (1). The independent variables are the fortnight periods, win margin, auction- and firm-fixed effects, political connection dummy, as well as the (log) number of firms' employees in the −sixth fortnight. The estimates are calculated using the package ("lfe," Gaure 2013) and show the LATE, difference in employment growth rates between winners and runner-ups. The point estimates and standard errors are transformed to absolute employment increases based on the coefficients and the mean number of employees (given in "mean employees") in the −sixth fortnight.
***, **, and * Significant at the 1%, 5%, and 10% level.

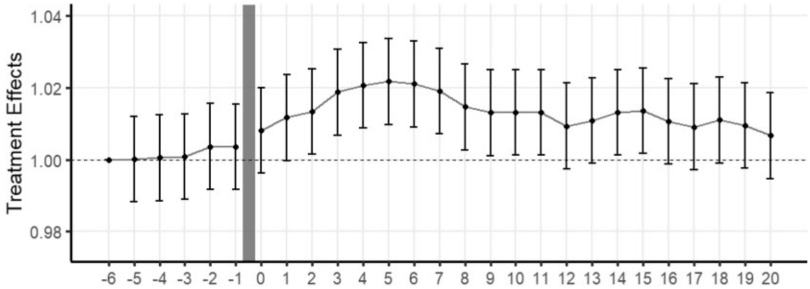

Figure 5. The longer-term impact of PPC on firm employment.

Notes: The point estimates with confidence intervals at 90% are extracted from the main model (Table 8). The sample is in close auctions with a difference between winners and runners-up up to 10%.

where the main difference is the length of the panel, which goes to fortnight 20 in Equation (2). The results in Figure 5 (based on Table 8) show that the PPC effect on firms' employment is persistent after the last period of the short-run estimation in Equation (1) but starts to fade out from the 16th fortnight onward (from day 238). Thus, we find no evidence for the longer persistence of the effects.

Runners-up in close auctions are active in public procurement and got close to winning, so they are in a good position to grow despite the loss of the auction. To investigate mechanisms for no longer persistence, we further investigate the Matthew effect and substitution of the public for the private market.

5.2. 1 Matthew effect

The Matthew effect of accumulated advantage (short: Matthew effect) suggests that the rich get richer and the poor get poorer. In our study setting, this could be interpreted as those firms that closely win a PPC get more and more PPCs (see Ferraz et al. 2021). The Matthew effect is tested with two models. In both models, the sample of close winners and runners-up is used, and at each auction time t, the number of PPCs received and their value are calculated for each winner or runner-up in 30, 90, 180, 270, and 360 days after the auction j at time t. Tables 9 and 10 show, contrary to the hypothesized, that the winners receive a lower number of PPCs and a lower value of PPCs in the next year. This can be interpreted with two competing arguments. On the positive side, firms are working at their full capacity and are not interested in acquiring new PPCs; alternatively (on the negative side), future research could look into potential circles of bid rigging, whereby a firm wins a PPC and then lets other firms win the next ones (i.e., they switch in a circle) in exchange for lower competition.

5.2. 2 Substitution

As shown in Section 5.2.1, if a runner-up loses an auction, it can offset a loss by winning the next auctions (Table 9). Therefore, during a longer period, for example, a year, the winner and runner-up could obtain a similar auction value regardless of the close auctions. Two points emerge.

First, while this is a lesser issue in our study context because the dependent variable is measured at the daily level, closely winning or losing a PPC should be an important event for the firm that is not easily offset if it is to provoke a positive firm employment reaction. To shed light on this topic, the total value of PPCs won for each firm-year is calculated. Panel (a) of Figure A6 shows a jump in the total yearly public amount won among the winners compared with the runners-up. Tables 11 and 12 show a statistically significant positive difference

Table 8. Long-term effects on employment.

	Main (within 10%) (1)	Within 8% (2)	Within 6% (3)	Within 4% (4)	4–0.5% (5)
−5	0.013	0.067	−0.076	0.042	0.203
−4	0.064	0.055	−0.030	0.153	0.391
−3	0.106	0.047	−0.082	0.123	0.258
−2	0.453	0.403	0.161	0.430	0.302
−1	0.440	0.303	0.042	0.380	0.358
0	1.012	0.834	0.471	0.808	0.846
1	1.454	1.408	0.898	0.885	0.962
2	1.661*	1.689*	1.308	1.234	1.345
3	2.333**	2.301**	2.110**	2.129*	2.069*
4	2.569***	2.378**	2.171**	2.538**	2.602**
5	2.703***	2.474***	2.175**	2.630**	2.998**
6	2.618***	2.464***	2.109**	2.701**	2.933**
7	2.367***	2.197**	1.907*	2.820**	2.979**
8	1.822**	1.721*	1.435	2.496**	2.489**
9	1.622*	1.378	1.249	1.960*	1.856
10	1.629*	1.126	0.951	1.447	1.225
11	1.636*	1.061	0.862	1.361	0.877
12	1.164	0.540	0.407	0.952	0.303
13	1.346	0.617	0.611	0.989	0.270
14	1.641*	0.801	0.703	1.192	0.676
15	1.684*	0.962	1.022	1.570	0.838
16	1.316	0.531	0.679	1.436	0.540
17	1.133	0.239	0.274	0.875	−0.156
18	1.369	0.262	0.460	1.332	0.523
19	1.175	0.094	0.658	1.536	0.859
20	0.830	−0.393	0.489	1.287	0.731
Mean employees	124.2909	120.9615	122.5415	123.4315	122.0516
N	71,712	62,370	49,302	36,315	29,997
R^2	0.622	0.618	0.643	0.645	0.644
Adjusted R^2	0.587	0.582	0.608	0.609	0.608
Residual Std. Error	12.193 (df = 65,699)	11.650 (df = 57,033)	10.941 (df = 44,963)	10.660 (df = 33,004)	10.699 (df = 27,199)

Notes: The dependent variable is employment growth at firm–auction level in each fortnight period (from −6 to 20). The model is estimated with Equation (1). The independent variables are the fortnight periods, win margin, auction- and firm-fixed effects, political connection dummy, as well as the (log) number of firms' employees in the −sixth fortnight. The estimates are calculated using the package ("lfe," Gaure 2013) and show the LATE, difference in employment growth rates between winners and runner-ups. The point estimates and standard errors are transformed to absolute employment increases based on the coefficients and the mean number of employees (given in "mean employees") in the −sixth fortnight.
***, **, and * Significant at the 1%, 5%, and 10% level.

between winners and runners-up in the PPC value won.[37] Second, even if there is a difference in the value of PPCs received between winners and runners-up, these differences might be substituted by working more for the private market. The value of PPCs from the firm's total revenue is subtracted to obtain an estimate of the firm's market revenue. The change in market revenue is less obvious. A model is run (in Tables 11 and 12) to analyze the difference in means of public and market revenue between the winners and runners-up. Tables 11 and 12 show a statistically significant negative difference in the market revenue. The jump in public revenue is larger than the change in market revenue (Panel a versus b, Figure A6).

[37] Natural logs of PPC value to decrease the influence of very large PPCs are used.

Table 9. Effect of winning a close auction on future auction victories.

	Within 30 days		**Within 90 days**		**Within 180 days**		**Within 270 days**		**Within 360 days**	
	Multiple bid (1)	**Single bid (2)**	**Multiple bid (3)**	**Single bid (4)**	**Multiple bid (5)**	**Single bid (6)**	**Multiple bid (7)**	**Single bid (8)**	**Multiple bid (9)**	**Single bid (10)**
Close winner	–0.059*	0.004	–0.206***	0.010	–0.344***	0.033	–0.457***	0.009	–0.466***	0.003
(within 10%)	(0.033)	(0.015)	(0.057)	(0.027)	(0.076)	(0.033)	(0.090)	(0.039)	(0.105)	(0.045)
N	3268	3268	3268	3268	3268	3268	3268	3268	3268	3268
R^2	0.306	0.308	0.511	0.433	0.662	0.626	0.739	0.693	0.772	0.738
Adjusted R^2	0.092	0.094	0.360	0.258	0.557	0.510	0.658	0.599	0.702	0.657
Residual Std. Error (df = 2495)	0.808	0.359	1.395	0.655	1.876	0.815	2.222	0.970	2.576	1.108

Notes: The dependent variable is the number of awarded PPC in a given period following a close auction victory. It is split by single- and multiple-bidder auctions and by five time periods. Its independent variable is a dummy (which is equal to 1 if the bidder is a victor only in a close auction, and 0 if the bidder is a runner-up in a close auction). The control variable is unique firm IDs (OIB).
***, **, * Significant at the 1%, 5%, and 10% level.

Table 10. Effect of winning a close auction on future PPC awarded (natural log) value.

	Within 30 days		Within 90 days		Within 180 days		Within 270 days		Within 360 days	
	Multiple bid (1)	Single bid (2)	Multiple bid (3)	Single bid (4)	Multiple bid (5)	Single bid (6)	Multiple bid (7)	Single bid (8)	Multiple bid (9)	Single bid (10)
Close winner (within 10%)	–0.546**	0.091	–0.843***	0.054	–0.743***	0.331	–0.937***	0.262	–0.973***	–0.035
	(0.256)	(0.159)	(0.253)	(0.211)	(0.223)	(0.214)	(0.202)	(0.215)	(0.185)	(0.211)
N	3268	3268	3268	3268	3268	3268	3268	3268	3268	3268
R^2	0.310	0.298	0.492	0.424	0.605	0.564	0.661	0.606	0.690	0.650
Adjusted R^2	0.096	0.081	0.334	0.245	0.482	0.429	0.556	0.484	0.594	0.542
Residual Std. Error (df = 2495)	6.299	3.914	6.233	5.202	5.505	5.276	4.963	5.289	4.546	5.191

Notes: The dependent variable is the natural log value of awarded PPC in a given period following a close auction victory. It is split by single- and multiple-bidder auctions and by five time periods. Its independent variable is a dummy (which is equal to 1 if the bidder is a victor only in a close auction, and 0 if the bidder is a runner-up in a close auction). The control variable is unique firm ids (OIB).
***, **, and * Significant at the 1%, 5%, and 10% level.

Table 11. Effects of winning a PPC on the market and public revenue: close auction sample.

	Period $t-1$		Period t		Period $t+1$	
	Public revenue (1)	Market revenue (2)	Public revenue (3)	Market revenue (4)	Public revenue (5)	Market revenue (6)
PPC win	−0.087	0.021	0.546***	−0.057***	−0.288**	0.006
	(0.092)	(0.015)	(0.075)	(0.013)	(0.117)	(0.010)
N	3058	2602	3058	2718	3058	1468
R^2	0.919	0.973	0.917	0.980	0.877	0.996
Adjusted R^2	0.887	0.960	0.883	0.971	0.827	0.993
Residual Std.	2.119	0.315	1.726	0.270	2.684	0.139
Error	(df = 2179)	(df = 1798)	(df = 2179)	(df = 1906)	(df = 2179)	(df = 865)

Notes: OLS models on the subsample of close auction within 10% win margin. Both dependent variables, the public revenue and the market revenue, are in natural logs. The main independent variable is a dummy indicating whether a firm is a winner or runner-up in an auction. Period $t-1$ is the accounting year before the year of auction result, t is the year of auction results, and $t+1$ year after. The unit of observation is firm–auction. All models include firm-fixed effects and a control variable for firm size (number of employees).
***, **, and * Significant at the 1%, 5%, and 10% level.

Table 12. Effects of winning a PPC on growth in market and public revenue: close auction sample.

	Period (t)−period $(t-1)$		Period $(t+1)$−period $(t-1)$	
	Public revenue (1)	Market revenue (2)	Public revenue (3)	Market revenue (4)
PPC win	0.634***	−0.070***	−0.200	−0.014
	(0.131)	(0.023)	(0.166)	(0.015)
N	3058	2411	3058	1388
R^2	0.808	0.787	0.762	0.967
Adjusted R^2	0.730	0.689	0.666	0.943
Residual Std.	3.007	0.463	3.799	0.203
Error	(df = 2179)	(df = 1659)	(df = 2179)	(df = 812)

Notes: OLS models on the subsample of close auction within 10% win margin. Both dependent variables, the public revenue and the market revenue, are calculated as the difference in natural logs between periods t and $t-1$ and t + 1 and $t-1$. Period $t-1$ is the accounting year before the year of auction result, t is the year of auction results, and $t+1$ year after. The unit of observation is firm–auction. The main independent variable is a dummy indicating whether a firm is a winner or runner-up in an auction. All models include firm-fixed effects and a control variable for firm size (number of employees).
***, **, and * Significant at the 1%, 5%, and 10% level.

The above suggests that runners-up substitute public revenue for market revenue when they lose a PPC in a close auction.[38] This increase in the market revenue of runners-up, however, is not enough to catch up to the winners' new PPC value in the current year. In other words, runners-up lose larger PPCs in close auctions, which they partly offset by working more on the market and by winning more PPCs of smaller size.

5.3 Dose–response functions

We estimate the dose–response functions within the close auction sample. To do so, we split the sample based on PPC relative size into four dosages (bins). The relative PPC size is

[38] After losing the PPC in a close auction and winning the next PPCs, runners-up still received on average €330,000 fewer public funds. A detailed calculation of the difference in PPC value between the PPC lost by the runner-up and the future PPCs is available to readers upon request.

defined as the PPC value divided by the firm's last year's total revenue. Four bins are divided based on the quantiles of the relative PPC size distribution. The first dose is defined as relative PPCs up to 2.6%, the second dose above 2.6 up to 7.3%, the third dose above 7.3 up to 19.8%, and the fourth dose above 19.8%. In the first dose, the firms are on average larger (at 191 employees) and are being awarded relatively small contracts (median at €146,047). In the second dose, firms have 118 employees on average and are allocated somewhat larger contracts (median at €203,085). The third dose includes firms with 106 employees on average winning larger contracts (median at €313,417). The final fourth dose includes smaller firms with 82 employees on average being awarded larger contracts (median at €641,412).

Equation (1) is run on the four subsamples, taking the winner and runner-up from each auction within a particular dose. Focusing on the fifth fortnight of Table 13, the results show no evidence for a positive impact of the first two PPC doses (below 7.3%) on firms' employment (Figure 6). Such firms probably can conduct the awarded public contracts with existing capacities. The third dose yields a positive impact on the firms' employment in the magnitude of 2.3 [±0.9]. Finally, the fourth dose yields a positive impact on employment of the highest magnitude among the four dosages (3.5 [±0.9]).

The lack of evidence collected for the impact of the small doses can partly explain the high costs per job created (and the small magnitude of the effect). However, while the largest PPC dose shows the largest absolute impact on firms' employment, the cost per job is higher than the main estimates (at €184,844; Table A24) which is larger than the estimates in the United States ($120,000; Wilson 2012). Therefore, the small magnitude is not explained by the PPC size.

5.4 Outsourcing practice

Winning firms can fulfill the contractual work obligations in-house, but can also decide to outsource parts of the PPC outside the firm. These activities are usually conducted as a cost-cutting measure or because specialized outside expertise is needed for the project. If firms do outsource, they usually conduct part of the contractual activities in-house and part is outsourced (Arvanitis and Loukis 2013). The two most usual forms of outsourcing are: firms outsource parts of the PPC to other firms and firms 'borrow' workers from a firm (i.e., agency workers) specialized for construction work.[39] We estimate heterogeneous effects with respect to the (i) intensity of outsourcing construction works to its (sub)suppliers and (ii) firms' intensity of using agency workers. We use detailed financial statements of all firms in the sample to construct (i) the ratios of outsourcing costs in the firm's total costs and (ii) the ratios of external labor costs to total labor costs.

The sample of firms in close auctions is split into quantiles based on the (i) ratio of outsourcing costs in total firm costs and based on (ii) the ratio of agency workers' costs to total labor costs. Logically, having high ratios of outsourcing would lead to a lower estimated impact because the impact spills over into the production network. Similarly, external labor is not registered in the database, and thus the unbiased impact of PPC on firms' employment should be in the subsample of firms with low usage of agency workers. We estimate Equation (1) on the four samples.

Results (Table 14) confirm that the effects are concentrated in those firms with low usage of external labor, but also show that the effect is of higher magnitude when the majority of all work is conducted *in-house*. Figure 7 shows the magnitude in the close auction sample when construction firms have a low intensity of outsourcing. In particular, the PPC impact

[39] Thus, employees are not registered within the winning firm but are "borrowed" from a specialized firm for such agency workers.

Table 13. The impact of PPC on firms' employment by relative PPC size.

	Main (1)	Below 2.6% (2)	2.6–7.3% (3)	7.3–19.8% (4)	Above 19.8% (5)
−5	0.013	−0.050	0.354	0.176	−0.147
	(0.572)	(1.742)	(1.098)	(0.942)	(0.908)
−4	0.064	−0.166	−0.125	0.357	0.134
	(0.572)	(1.742)	(1.098)	(0.942)	(0.908)
−3	0.106	−0.073	0.006	0.404	0.072
	(0.572)	(1.742)	(1.098)	(0.942)	(0.908)
−2	0.453	−0.374	0.412	0.612	0.524
	(0.572)	(1.742)	(1.098)	(0.942)	(0.908)
−1	0.440	−0.695	0.438	0.459	0.431
	(0.572)	(1.742)	(1.098)	(0.942)	(0.908)
0	1.012*	−0.810	1.033	0.689	1.207
	(0.572)	(1.742)	(1.098)	(0.942)	(0.908)
1	1.454***	0.089	1.603	0.480	1.588*
	(0.572)	(1.742)	(1.098)	(0.942)	(0.908)
2	1.661***	−0.398	0.963	0.826	2.296***
	(0.572)	(1.742)	(1.098)	(0.942)	(0.908)
3	2.333***	−0.002	1.442	1.587*	2.789***
	(0.572)	(1.742)	(1.098)	(0.942)	(0.908)
4	2.569***	−0.028	1.380	1.921**	3.029***
	(0.572)	(1.742)	(1.098)	(0.942)	(0.908)
5	2.703***	−0.222	0.879	2.348**	3.474***
	(0.572)	(1.742)	(1.098)	(0.942)	(0.908)
Mean employees	124.2909	191.1044	117.6022	106.3483	82.2878
N	31,872	7092	7572	7752	7380
R^2	0.629	0.682	0.624	0.614	0.600
Adjusted R^2	0.544	0.596	0.524	0.512	0.494
Residual Std. Error	7.563 (df = 25,934)	9.035 (df = 5590)	6.971 (df = 5984)	6.436 (df = 6121)	6.206 (df = 5835)

Notes: Column (1) shows the estimates for the main sample of auctions within 10%. Columns 2–5 show results for PPC auctions depending on the final PPC value and the mean number of employees at the beginning of the −sixth fortnight and pre-defined dosages. The dependent variable is employment growth at firm–auction level in each fortnight period (from −6 to 5). The model is estimated with Equation (1). The independent variables are the fortnight periods, win margin, auction- and firm-fixed effects, political connection dummy, as well as the (log) number of firms' employees in the −sixth fortnight. The estimates are calculated using the package ("lfe," Gaure 2013) and show the LATE, difference in employment growth rates between winners and runner-ups. The point estimates and standard errors are transformed to absolute employment increases based on the coefficients and the mean number of employees (given in "mean employees") in the −sixth fortnight.
***, **, and * Significant at the 1%, 5%, and 10% level.

on firm employment is 3.7 (±0.9) employees.[40] We take this magnitude as the more accurate estimate of the PPC impact on firms' employment. With the new magnitude and the median PPC amount size,[41] we quantify the public cost per directly created job at €58,600 (€49,800–€65,100) VAT included.

The magnitude of 3.7 (±0.9) employees is higher than the main model which yields a positive PPC impact on employment in the magnitude of 2.7 (±0.6) employees (i.e., Table 3). Importantly, while no evidence is found of a positive impact of PPC on the employment of those construction firms with a high share of outsourcing or high usage of agency workers, it does not directly imply there is no effect among these firms, but more probably, that the effect has spilled over to the winners' suppliers and higher usage of agency workers.

[40] The point estimate ranges from 2.9 to 3.7.
[41] €213,974.67 at the median, see Table A24.

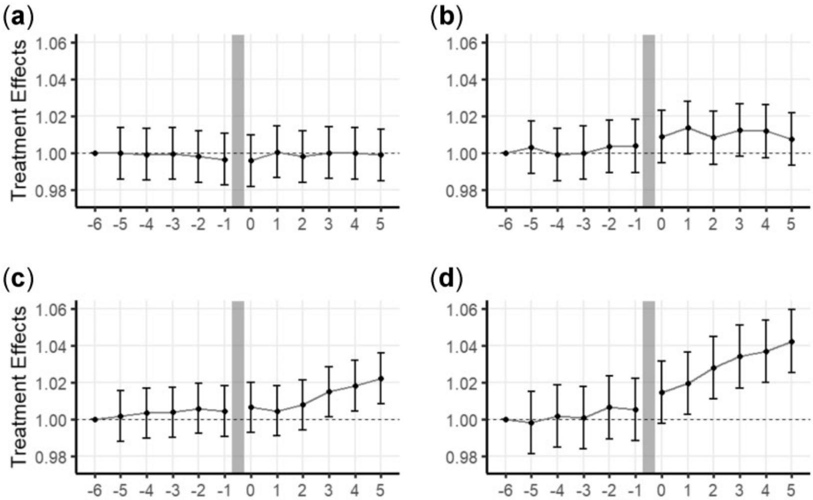

Figure 6. Dose–response functions: PPC impact on firm employment.

Notes: The point estimates with confidence intervals at 90% are extracted from the main model (Table 13). The sample is in close auctions with a difference between winners and runners-up up to 10%. The panels (**a**) is the first dose (up to 2.6%), (**b**) is the second dose (2.6–7.3%), (**c**) is the third dose (7.3–19.8%), and (**d**) is the fourth dose (above 19.8%).

6. CONCLUSIONS

Macroeconomics has gone a long way in estimating the effects of government spending on GDP growth. However, macroeconomics is based on microeconomics and micro-level effects of government spending are still under-researched. These micro gaps primarily include the effects of government spending on households and firms. We tackle the micro-level effects of government spending on the employment of firms supplying the government with construction works. We speak to the scarce literature on estimating the effects of PPCs on firms' job creation (Gugler et al. 2020; Hjort et al. 2020; Ferraz et al. 2021) and to the literature on political favoritism in the public procurement market (Szucs 2017). We examine PPC and firms, considering moral hazard issues between self-interested politicians and profit-maximizing firms.

To estimate the effects of PPC on firm employment, we exploit sealed bid auctions in PPCs of construction works, daily employment variation, and discontinuity in bidders' win margin. We show that a transparent and competitive auction procedure is associated with cost savings and fewer contracts awarded to politically connected firms. Training firms on how to compete for public contracts could increase auction competition; for example, Hjort et al. (2020) find that teaching firms how to compete for government contracts increases the quantity and quality of contracts won. Auction competition can also depend on the magnitude of public demand for construction works and the context (e.g., booming private demand). Notably, while politically connected firms are allocated a substantial share of contracts, winners and runners-up are, on average, not statistically different in political connections. We also examine information leakage and do not find compelling evidence for information leakage as a threat to the identification strategy. Therefore, any remaining difference in firms' employment growth is due to PPC.

Winning a PPC has a positive impact on a firm's short-run employment, with a relatively small initial estimated magnitude of the effect (2.7 employees [±0.6]) and, relatedly, higher initial cost per job directly created (at €93,223) than the majority of estimates for PPCs or public grants (Gugler et al. 2020; Dvouletý et al. 2021; Ferraz et al. 2021; Srhoj and Zilic 2021).

Table 14. Heterogeneous effects—firms' outsourcing costs as a share of total costs.

	Within 10% (main) (1)	First bin (0–19%) (2)	Second bin (19–30%) (3)	Third bin (30–41%) (4)	Fourth bin (41% +) (5)
−5	0.013	−0.078	−0.267	−0.448	0.742
	(0.572)	(0.923)	(0.780)	(1.383)	(1.303)
−4	0.064	−0.258	0.184	−0.822	0.485
	(0.572)	(0.923)	(0.780)	(1.383)	(1.303)
−3	0.106	−0.005	0.320	−1.010	−0.051
	(0.572)	(0.923)	(0.780)	(1.383)	(1.303)
−2	0.453	0.260	0.565	−0.547	0.285
	(0.572)	(0.923)	(0.780)	(1.383)	(1.303)
−1	0.440	0.347	1.102	−1.525	0.088
	(0.572)	(0.923)	(0.780)	(1.383)	(1.303)
0	1.012*	0.954	1.587**	−1.619	−0.014
	(0.572)	(0.923)	(0.780)	(1.383)	(1.303)
1	1.454**	1.530*	1.744**	−2.088	−0.145
	(0.572)	(0.923)	(0.780)	(1.383)	(1.303)
2	1.661***	2.098**	1.257	−1.786	0.070
	(0.572)	(0.923)	(0.780)	(1.383)	(1.303)
3	2.333***	2.907***	1.354*	−1.221	1.441
	(0.572)	(0.923)	(0.780)	(1.383)	(1.303)
4	2.569***	3.491***	1.606**	−1.402	1.517
	(0.572)	(0.923)	(0.780)	(1.383)	(1.303)
5	2.703***	3.651***	2.199***	−1.806	1.367
	(0.572)	(0.923)	(0.780)	(1.383)	(1.303)
Mean employees	124.2909	97.7724	100.5589	142.7472	132.3045
N	31,872	7668	7572	7584	7584
R^2	0.629	0.618	0.652	0.645	0.651
Adjusted R^2	0.544	0.504	0.546	0.539	0.546
Residual Std. Error	7.563 (df = 25,934)	6.199 (df = 5915)	5.196 (df = 5802)	8.387 (df = 5825)	8.237 (df = 5836)

Notes: We split the data into four similarly sized samples according to the bidder's share of outsourcing costs in the firm's total costs. The dependent variable is employment growth at firm–auction level in each fortnight period (from −6 to 5). The model is estimated with Equation (1). The independent variables are the fortnight periods, win margin, auction- and firm-fixed effects, political connection dummy, as well as the (log) number of firms' employees in the −sixth fortnight. The estimates are calculated using the package ("lfe," Gaure 2013) and show the LATE, difference in employment growth rates between winners and runner-ups. The point estimates and standard errors are transformed to absolute employment increases based on the coefficients and the mean number of employees (given in "mean employees") in the −sixth fortnight.
***, **, and * Significant at the 1%, 5%, and 10% level.

The small initial magnitude motivated further investigation of heterogeneity and mechanisms that could explain such a high cost per job in the short run. Contrary to Ferraz et al. (2021), our investigation of the longer-term effects finds evidence that the effects phase out in less than a year, suggesting the longevity of effects or later-increasing effects do not explain the lower short-term magnitude. However, we find four factors that can explain the lower magnitude. The first two relate to the counterfactual behavior of the runners-up. On the one hand, runners-up win more PPCs over the months after losing a close auction, which raises suspicion of bid rigging (Luz and Spagnolo 2017). However, these new PPCs won by the runners-up are smaller on average and do not provide enough public revenue for the runners-up to reach the winner's level of public revenue received. On the other hand, the context of our research is in the expansion period, and we find empirical support that runners-up engage in *substitution* of the public for market revenue, showing runners-up start working more with households and private firms (i.e., private demand). Third, we estimate

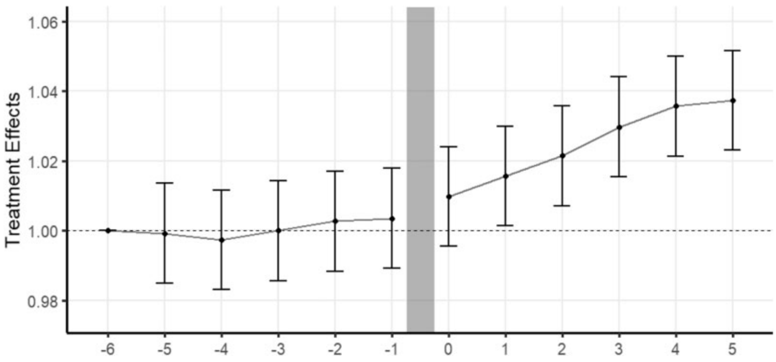

Figure 7. Heterogeneous effects: low outsourcing share.

Notes: The point estimates with confidence intervals at 90% are extracted from the main model (Table 14). The sample is in close auctions with a difference between winners and runners-up up to 10%.

dose–response functions and find non-linearity in the effects of PPC on firms' employment with respect to the relative PPC size (i.e., dosage). We find no evidence for a positive effect of relatively small PPCs on firms' employment and document positive effects for relatively larger dosages. While the impact increases in the third and fourth PPC size bins, the cost per job becomes more expensive in the fourth bin. Non-linear effects suggest that chopping the PPCs into relatively more minor contracts might not be beneficial for maximizing the effect on firms' employment. However, with a larger dosage, the effect per euro decreases after a threshold, which is why PPC size only partly explains the high cost per job created. Fourth, we document the heterogeneity of effects concerning firms' outsourcing decisions. In particular, the effect is concentrated in firms that conduct the majority of work themselves or firms that use a small share of agency workers. On average, firms employ 3% of the winning firms' workforce, 3.7 new employees per PPC, which gives the public cost per job directly created at €58,600 (€49,800–€65,100).

The anatomy of identified new hires shows the majority are male, hired on fixed-term contracts, and in job positions requiring lower education. Additionally, most new hires are working in the country for the first time from which a non-negotiable share is probably foreign workers. Previously employed new hires usually transfer from other construction firms or are experiencing a brief unemployment spell.

In sum, during an expansion period, we show that public spending on construction works creates short-term low-quality jobs in firms winning contracts, however, at a higher cost per job than other public policies (e.g. public grants for firms in the EU; Dvouletý et al. 2021).

Tackling all the interesting questions related to PPCs is beyond the scope of this study, but we suggest several future research streams. Future research could tackle the impact of PPC on the employment of winners' suppliers. In addition, investigating PPC effects on other production function variables, that is, inputs like capital, intermediate inputs, and technology, and outputs like value-added and sales, could yield exciting findings. Furthermore, researchers could estimate the effects of PPC on firms in non-construction sectors (e.g., manufacturing, retail, or service sectors), as well as the effects of a change in political power on bidders' behavior, resource allocation, competition, price markups in PPCs. Future research can investigate local and regional effects of increased government spending (e.g., Nakamura and Steinsson 2014), as well as the micro-effects of "mega" PPCs (e.g., significant highways or bridges [Holl 2016]). Finally, potential bid rigging activities require more analysis (e.g., Chassang et al. 2022). Extending our analysis to consider these aspects is an exciting yet challenging avenue for future research.

APPENDIX

APPENDIX A: POLITICAL CONNECTIONS AND PUBLIC PROCUREMENT CONTRACTS

From a standard three-sector circular flow model, the firms produce products demanded by households and other firms (i.e., private demand) and by public institutions (i.e., government, regional, local institutions, and SOEs, i.e., public demand). Moral hazard issues can occur between profit-maximizing firms and politicians who might not only represent the public interest but also their own (among others see Szucs 2017; Lehne et al. 2018; Baltrunaite 2020; Bosio et al. 2020; Vuković 2020; P. Andreyanov, A. Davidson, and V. Korovkin, unpublished data). To alleviate these moral hazard issues, governments around the world usually have some kind of transparent and competitive procedure, like auctions, to decide which firms will win the PPC. However, even though these procedures exist, their outcomes are rather heterogeneous. A recent macro-level study on a sample of 187 countries found a positive association between perceived country-level corruption and unfavorable procurement processes and outcomes (Bosio et al. 2020). Along this line, at the micro-level, Szucs (2017) shows that giving more discretion to public buyers increases the public costs and lowers the productivity of selected suppliers, Vuković (2020) suggests PPCs as a tool to buy votes and build a crony winning coalition, Lehne et al. (2018) suggest that politicians can adjust resource allocation in favor of their family members at a great cost to the society, and P. Andreyanov, A. Davidson, and V. Korovkin (unpublished data) and Baltrunaite (2020) suggest resource allocation can work via an information leakage channel between the contracting authority and profit-maximizing firms. These political economy studies suggest that selection into PPCs is not random, and it depends not only on which private firms best serve the public interest, but also the self-interest of politicians. Additionally, firms can get organized into cartels to do price fixing and bid rigging in order to overcharge the public for their services (Smuda 2014). These cartels are again fought not only by transparent procurement procedures, but also by fining the wrongdoers, providing leniency for wrongdoers that blow the whistle, and providing monetary awards to whistleblowers (Bigoni et al. 2012; Luz and Spagnolo 2017).

APPENDIX B: INFORMATION LEAKAGE CHANNEL: INTERVIEWS

To shed some light on information leakage theory, two experts were interviewed. First, a prominent university law professor—an expert on PPC, and second, PPC auctions information technology expert. Both agree that information leakage in the auction system is technically impossible without the consent of both sides, the contracting authority and the bidder. This is so because each bid is encrypted via the online auction system and has two keys (contracting authority key and bidder key). These safety protocols are certified by multiple certifying authorities, among which also European Union institutions. Thus, the only way to obtain information on a competitor's bid is to enter two keys in the system. In other words, the bidder would have to want to provide such information to other auction participants. Another option could be that it is not the information leakage, but bid rigging, where runners-up place artificial bids slightly more expensive than the winner. Two interviewed experts were asked whether they are aware of such collusive behavior, and both suggest this would be highly unusual bidding behavior, as collusion is probably more frequently conducted by not participating in the auction, which is a less costly approach given that participation requires labor hours worked.

APPENDIX C: TABLES

Table A1. Public procurement in the Republic of Croatia.

Year (1)	GDP (2)	PPC (3)	PPC in GDP (%) (4)	PPC—EOJN (5)	Simple PPC (6)	Number of PPC (7)	Number of PPC w. MEAT (%) (8)	Share of PPC value MEAT (9)	Construction of PPC's value in PPC's at EOJN (%) (10)
2015	339.7	40.6	11.9	31.1	9.5	15,485	2.4	7.8	46.9
2016	351.2	44.8	12.8	34.6	10.3	13,838	2.5	6.6	27.7
2017	366.4	40.5	11.0	31.0	9.4	11,408	57.1	57.7	42.0
2018	383.0	46.6	12.2	36.6	10.0	18,112	95.5	95.4	50.8

Notes: All monetary values are given in billion Kuna. Column (3) shows the total PPC value awarded. Column (4) shows the total PPC value awarded as % of GDP. Column (5) shows the value of PPCs published at EOJN. Column (6) gives the value of all PPC which do not legally require a tendering process (all PPC whose final value is under 250,000 HRK [with VAT]). Column (8) shows % of PPC awarded by MEAT criteria, Column (9) does the same but comparing values of PPC. Column (10) simply looks at what % of PPC at EOJN have their CPV start with 45*xy*.
Source: Statistical Reports on Public Procurement, link: http://www.javnanabava.hr/default.aspx?id=3425.

Table A2. Definition of non-political variables used in the analysis.

Category	Variable	Description
Employment	Employees	Number of employees within a firm (six fortnights prior to the auction results).
	Employees, HE	Number of employees with a higher education (HE) level within a firm (six fortnights prior to the auction results).
	Employees, LE	Number of employees with a lower education (LE) level within a firm (six fortnights prior to the auction results).
	Hires	Number of hires within a firm (per day, during the six fortnights period prior to the auction results).
	Fires	Number of fires within a firm (per day, during the six fortnights period prior to the auction results).
	Tenure	Number of days an average employee is employed in a firm (six fortnights prior to the auction results).
	Employee age	Age of an average employee (on the day he is employed) that is employed in a firm (six fortnights prior to the auction results).
	Non-fixed term contracts	Percentage of non-fixed term contracts within a firm (six fortnights prior to the auction results).
Projects	Backlog extensive	Number of government contracts won by a firm during the 3 years prior to an auction (includes contracts from whole 3 years prior).
	Backlog intensive	Value of government contracts won by a firm during the 3 years prior to an auction (includes contracts from the whole 3 years prior) VAT is not included.
	Distance to contracting authority	Distance between contractors' and firms' headquarters (air distance—in kilometers).
	Total assets	Total assets of a firm according to the nearest end-of-year financial reports prior to the auction results.
	Current assets	Current assets of a firm according to the nearest end-of-year financial reports prior to the auction results.

(continued)

Table A2. (continued)

Category	Variable	Description
	Fixed assets	Fixed assets of a firm according to the nearest end-of-year financial reports prior to the auction results.
	Total liabilities	Total liabilities of a firm according to the nearest end-of-year financial reports prior to the auction results.
Balance sheet	Non-current liabilities	Non-current liabilities of a firm according to the nearest end-of-year financial reports prior to the auction results.
Income statement	Revenue	Revenue of a firm according to the nearest end-of-year financial reports prior to the auction results.
	Market revenue	Revenue of a firm (according to the nearest end-of-year financial reports prior to the auction results) subtracted by the "Public revenue."
	Public revenue	Total value a firm won one year prior to the auction results within the PPCs within our database. VAT is not included.
	EBITDA	EBITDA of a firm according to the nearest end-of-year financial reports prior to the auction results.
	Profit	Profit of a firm according to the nearest end-of-year financial reports prior to the auction results.
	Depreciation	Depreciation of a firm according to the nearest end-of-year financial reports prior to the auction results.
	Interest paid	Interest paid of a firm according to the nearest end-of-year financial reports prior to the auction results.
	Productivity	Revenue of a firm according to the nearest end-of-year financial reports prior to the auction results over the number of employees within a firm six fortnights prior to the auction results.
	Wage costs	Wage costs of a firm according to the nearest end-of-year financial reports prior to the auction results.
Financial ratios	EBITDA over assets	EBITDA over total assets of a firm according to the nearest end-of-year financial reports prior to the auction results.
	Profit over assets	Profit after tax over total assets of a firm according to the nearest end-of-year financial reports prior to the auction results.
	Debt ratio	Total liabilities over total assets of a firm according to the nearest end-of-year financial reports prior to the auction results.
	LR liabilities over assets	Non-current liabilities over total assets of a firm according to the nearest end-of-year financial reports prior to the auction results.
	Outsourcing over total expenses	Total outsourcing (external-work) costs over the total firms costs of a firm according to the nearest end-of-year financial reports prior to the auction results.
	External labor over total labor costs	Total cost of student workers, agency workers, subcontractors, and other one-off contractors over the total worker expenses of a firm according to the nearest end-of-year financial reports prior to the auction results.

(continued)

Table A2. (continued)

Category	Variable	Description
Education	(HE) Higher education level	Requirements for a specific job vacancy, containing: specialized doctorate degree, doctorate degree, master's degree (and specialized masters programs), bachelor's degree.
	(LE) Lower education level	Requirements for a specific job vacancy, any degree of education below the/not mentioned in the degrees required for an HE classification.
	Win margin	$\frac{\lvert \text{2nd best bids value} - \text{winning bids value} \rvert}{\text{2nd best bids value}}$
	Dependent variable	We calculate each auction participant's daily number of employees, starting from employment at day −84 (relative to the day of the auction awards), up to day +84. To simplify the model, the mean number of employees is calculated for each fortnight. Then, we take logs of each fortnight's mean number of employees.

Table A3. Political connection dummies.

Match	Name	Dummies, equal to 1 if:
	Donators	A firm has ever donated to any political party according to our database of donations.
Last name match	Reg. conn. (out of power)	A firm has, within its management/owners, anyone with the same last name as an ex-regional politician (substitutes of-, governors, member of county councils) in the county of the firm's headquarters.
	Reg. conn. (in power)	A firm has, within its management/owners, anyone with the same last name as a current regional politician (substitutes of-, governors, member of county councils) in the county of the firm's headquarters.
	Loc. conn. (out of power)	A firm has, within its management/owners, anyone with the last name as an ex-member of the local political representatives in the municipality of the firm's headquarters.
	Loc. conn. (in power)	A firm has, within its management/owners, anyone with the same last name as a current member of the local political representatives in power in the municipality of the firm's headquarters.
Full name match	GONG/Nat. conn.	A firm has, within its management/owners, anyone with the same full name as an ex- or a current member of the national parliament or the parties' representatives.
Final dummies	Any	Any of the previous dummies are 1.
	Any (robust surname)	If any of the excluded last name dummies are 1 (all previous dummies are recalculated to exclude the top 10 most frequent last names in each part of the state).
	Any (second order)	Any of the managers/owners within firms where Any = 1 are members of the management/owners.

Notes: "(in power)" refers to the ruling party in the observed part of the state. The elections on the regional and local levels took place in 2013 and 2017, meaning we consider a politician "in power" if he was a mayor/deputy mayor/member of the local council.

Table A4. Single bid auctions.

	Year	2016	2017	2018	Total
(1) All PPC	Count	1412	1449	1378	4239
(2) Single bid	Count	332	286	32	650
(3) Single bid	Share	0.2351	0.1974	0.0232	0.1533
(4) Single bid	Amount (in mil. €)	142.6631	322.7729	24.6659	490.1019
(5) Winners donating	Share	0.1754	0.1365	0.0498	0.1435
(6) Winners pol. conn.	Share	0.5367	0.7683	0.3626	0.6804
(7) Suspicious winners	Share	0.1049	0.4696	0.3882	0.3593
(8) Ad 5, 6, and 7 Won	Amount (in mil. €)	100.5075	285.1328	19.4697	405.11

Notes: In total, 4239 auctions are in the entire sample when we exclude only the auctions for which we do not have the necessary data. Of those 4239, 650 were single bid auctions, however, 332 more are excluded, as they become single bid auctions because of invalid and/or excluded bids. Those 332 are not observed in single bid auctions above, they are represented below. VAT is included.

Table A5. Single bid auctions by exclusion.

	Year	2016	2017	2018	Total
(1) All PPC	Count	1412	1449	1378	4239
(2) Single bid	Count	52	68	212	332
(3) Single bid	Share	0.0368	0.0469	0.1538	0.0783
(4) Single bid	Amount (in mil. €)	42.1524	225.2118	180.6239	447.9881
(5) Winners donating	Share	0.106	0.0805	0.228	0.1424
(6) Winners pol. conn.	Share	0.6498	0.9132	0.6713	0.7909
(7) Suspicious winners	Share	0.2479	0.8	0.1095	0.4696
(8) Ad 5, 6, and 7 Won	Amount (in mil. €)	30.4392	213.5094	139.1027	383.0513

Notes: Rows 5–7 show the share of values of PPC won by each of the groups of single bidders, respectively. Row 5 shows the share of the total value awarded to single bidders with previous donations to a political party, row 6 to single bidders with political connections (see Section 4.5.3), and row 7 to single bidders which are deemed suspicious (firms formed 1 year or sooner before the auction, firms with no employees, and firms which won an auction that surpassed 70% of their last year's revenue). The last row shows the value that was awarded to firms in rows 5–7 (overlap is accounted for). VAT is included.

Table A6. Multiple bid auctions.

	Year	2016	2017	2018	Total
(1) All PPC	Count	1412	1449	1378	4239
(2) Multiple bid	Count	1028	1095	1134	3257
(3) Multiple bid	Share	0.728	0.7557	0.8229	0.7683
(4) Multiple bid	Amount (in mil. €)	921.6257	732.3651	1137.792	2791.7828
(5) Winners donating	Share	0.1811	0.1423	0.1035	0.1393
(6) Winners pol. conn.	Share	0.364	0.5113	0.2946	0.3744
(7) Suspicious winners	Share	0.3698	0.2929	0.309	0.3249
(8) Ad 5, 6, and 7 Won	Amount (in mil. €)	677.078	523.1096	615.3558	1815.5434

Notes: We observe the 3257 auctions which had multiple valid bids. Of those, we, later on, exclude ones in which a winner or a runner-up is a firm for which we do not have the necessary employment data. We are left with 2859 auctions afterward. VAT is included.

Table A7. Bidder quantity effect on winning bid (as % of beginning estimate).

		Subsamples				
	All	**1 versus 2**	**1 versus 3**	**1 versus 4**	**1 versus (5–8)**	**1 versus (>8)**
	(1)	(2)	(3)	(4)	(5)	(6)
Two valid bids	−0.073***	−0.073***				
	(0.008)	(0.009)				
Three valid bids	−0.119***		−0.125***			
	(0.008)		(0.009)			
Four valid bids	−0.147***			−0.150***		
	(0.010)			(0.010)		
(Five–eight) valid bids	−0.178***				−0.176***	
	(0.009)				(0.009)	
(more than eight) valid bids	−0.277***					−0.258***
	(0.022)					(0.023)
Mean beginning estimate	0.78	0.77	0.66	0.57	0.89	1.16
N	4108	1971	1737	1446	1663	995
R^2	0.167	0.101	0.156	0.171	0.240	0.191
Adjusted R^2	0.156	0.079	0.133	0.144	0.218	0.151
Residual Std. Error	0.173 (df = 4055)	0.181 (df = 1922)	0.175 (df = 1690)	0.177 (df = 1399)	0.173 (df = 1615)	0.177 (df = 948)

Notes: We observe the entire sample of 4239 auctions, however we exclude 131 auctions as their winning bid (as % of the beginning estimate) is in the top or bottom 1% of observations. Meaning we observe 4108 auctions in Column (1), and their subsamples in other columns. Column (2), for example, regresses the ratio on the subsample of auctions that had either one or two valid bids, and the other remaining columns follow the same principle. Regression is controlled for the county, season, year, and any political connections and four digit CPV-specific effects. The mean beginning estimate is in mil. €.
***, **, and * Significant at the 1%, 5%, and 10% level.

Table A8. Political connections.

		Last name match				**Dummies**		
	GONG/ Nat. conn.	**Reg. conn. (out of power)**	**Reg. conn. (in power)**	**Loc. conn. (out of power)**	**Loc. conn. (in power)**	**Any**	**Any (with safe surnames)**	**Any (second order)**
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
All unique firms	1071	1071	1071	1071	1071	1071	1071	1071
All pol. connected firms	141	166	179	202	195	373	310	675
Share of pol. connected firms	0.13	0.16	0.17	0.19	0.19	0.35	0.29	0.64

Notes: The first five columns show statistics for any connection to politicians using a dummy of 1 for a full name match or a last name match. Columns (6)–(8) give an overview of all political connections anytime. For a more detailed explanation of the variables, see Table A3.

Table A9. Politically connected winners and competition.

	Political connection		Political connection (robust surname)		Suspicious firms	
	(1)	(2)	(3)	(4)	(5)	(6)
Dummy 2+ bidders	−0.067***		−0.039***		−0.024***	
	(0.015)		(0.014)		(0.009)	
Bids: 2		−0.055***		−0.016		−0.028***
		(0.019)		(0.017)		(0.011)
Bids: 3		−0.053***		−0.024		−0.038***
		(0.020)		(0.018)		(0.011)
Bids: 4		−0.073***		−0.062***		−0.012
		(0.022)		(0.021)		(0.013)
Bids: over 4		−0.095***		−0.074***		−0.011
		(0.020)		(0.019)		(0.011)
Distance to	−0.00005	−0.0001	−0.0001	−0.0001	−0.00000	0.00000
contracting authority	(0.0001)	(0.0001)	(0.0001)	(0.0001)	(0.0001)	(0.0001)
Log employees	0.138***	0.138***	0.160***	0.160***	−0.031***	−0.031***
	(0.005)	(0.005)	(0.004)	(0.004)	(0.003)	(0.003)
Observations	4057	4057	4057	4057	4057	4057
R^2	0.531	0.531	0.569	0.571	0.270	0.272
Adjusted R^2	0.449	0.450	0.495	0.496	0.144	0.145
Residual Std. Error	0.371 (df = 3458)	0.371 (df = 3455)	0.348 (df = 3458)	0.347 (df = 3455)	0.211 (df = 3458)	0.211 (df = 3455)
F-Statistic	6.538***	6.519***	7.641***	7.646***	2.141***	2.144***
	(df = 598; 3458)	(df = 601; 3455)	(df = 598; 3458)	(df = 601; 3455)	(df = 598; 3458)	(df = 601; 3455)

Notes: Dependent variables are taking value of 1 if the auction winner is politically connected and 0 otherwise. The model is a linear probability model. The regression also controls for the month and year of the auction awards, the municipality of the bidder, as well as the municipality of the contracting authority.
***, **, and * Significant at the 1%, 5%, and 10% level.

Table A10. Procurement contracts distribution by six-digit CPV.

Six-digit CPV	CPV description	Number of auctions	Number of share	Estimated value	Estimated values share	Final value	Final values share	Mean final value	Median final value
452331	Works on building highways and roads	614	0.14	447.08	0.13	356.11	0.12	0.58	0.18
450000	Building (unspecified)	269	0.06	225.41	0.06	182.79	0.06	0.68	0.21
454540	Reconstruction and renovation	201	0.05	117.36	0.03	97.25	0.03	0.48	0.2
452313	Works on constructing water and sewer pipelines	191	0.05	228.83	0.07	212.03	0.07	1.11	0.26
452330	Construction works, works on building foundations, and works on constructing surface highway roads	165	0.04	154.72	0.04	112.3	0.04	0.68	0.17
452310	Works on constructing pipelines, communication, energy, and water supply	143	0.03	71.92	0.02	61.98	0.02	0.43	0.17
452332	Different works on surface layer	136	0.03	43.59	0.01	35.94	0.01	0.26	0.12
454531	Maintenance	128	0.03	92.9	0.03	74.68	0.03	0.58	0.13
452000	Works on buildings or parts of high-rise and low-rise buildings	122	0.03	84.59	0.02	68.32	0.02	0.56	0.16
452321	Works on water supply pipelines	93	0.02	53.85	0.02	40.09	0.01	0.43	0.19
	In top 10	2062	0.49	1520.25	0.44	1241.49	0.42	0.6	0.19
	Total	4239	1.00	3476.19	1.00	2983.9	1.00	0.7	0.18

Notes: All values are given in mil. €. The CPV distribution encompasses all 4239 auctions in the sample.

Table A11. Auction summary.

	All auctions			Close auctions (within 10%)			Close auctions (within 6%)			Close auctions (within 4%)		
	Mean	St. Dev.	Median	Mean	St. Dev.	Median	Mean	St. Dev.	Median	Mean	St. Dev.	Median
Auction estimate	808.42	5957.64	220	902.82	5413.83	237.56	982.62	6373.25	233.33	923	6618.41	240
Winning bid	721.42	6467.31	178.51	804.7	5182.35	206.29	887.15	6121.14	203.4	832.5	6332.67	212.97
(Winning bid/auction estimate)	0.83	0.25	0.81	0.88	0.28	0.88	0.89	0.29	0.9	0.9	0.32	0.91
Runner-up bid	826.3	7672.3	209.88	837.57	5279.28	213.21	911.32	6222.02	209.45	844.13	6361.98	214.59
(Runner-up bid—winning bid)	104.88	1391.39	20.53	32.87	165.72	7.08	24.16	174.47	4.58	11.63	46.54	3.31
(Run.-up bid—win. bid)/ runner-up bid	0.1389	0.1657	0.0994	0.0423	0.0289	0.0388	0.0262	0.0175	0.0241	0.018	0.0118	0.016
Number of bidders	4	2.01	3	4.31	2.19	4	4.38	2.24	4	4.43	2.26	4

Notes: All monetary values are in thousands of euro. VAT is excluded. We observe only auctions for which both the employee and financial data are available.

Table A12. Auction distribution by geographic region of contracting authority.

Geographic region of contracting authority	All auctions				Close auctions (within 10%)				Close auctions (within 6%)				Close auctions (within 4%)			
	Auctions		Value		Auctions		Value		Auctions		Value		Auctions		Value	
	No.	Share	Sum	Share	No.	Share	Sum	Share	No.	Share	Sum	Share	No.	Share	Sum	Share
Dalmatia	380	0.13	247.95	0.12	212	0.15	132.84	0.11	133	0.13	97.94	0.11	94	0.13	38.75	0.06
City of Zagreb	1228	0.43	1243.96	0.6	582	0.41	722.54	0.63	401	0.4	563.86	0.64	298	0.4	408.31	0.67
Istria, Kvarner, Gorski Kotar, and Lika	420	0.15	200.1	0.1	222	0.15	117.48	0.1	163	0.16	78.79	0.09	122	0.17	61.27	0.1
Central Croatia (w/o City of Zagreb)	434	0.15	191.82	0.09	221	0.15	84.83	0.07	152	0.15	63.04	0.07	117	0.16	40.92	0.07
Slavonia	396	0.14	178.62	0.09	199	0.14	97.99	0.08	145	0.15	78.3	0.09	106	0.14	64.42	0.1
Number of auctions	2859	1	2062.45	1	1436	1	1155.68	1	994	1	881.93	1	737	1	613.67	1
Number of unique firms	1071				824				719				634			

Notes: All monetary values are in millions of euro. VAT is excluded. We observe only auctions for which both the employee and financial data are available.

Table A13. Bid timing.

	Dummy	Time difference (in hours)		
		Total	Dummy = 0	Dummy = 1
Min.	0.00	–287.57	0.02	–287.57
First Qu.	0.00	–17.71	0.84	–24.00
Median	0.00	0.02	5.67	–17.94
Third Qu.	1.00	5.67	24.00	–1.17
Max.	1.00	432.00	432.00	–0.00
Mean	0.50	–2.59	27.47	–33.00

Notes: The table shows the bid timing data on a sample of close bids for which the exact receival time of each bid was available. We examine 268 auctions in 2018, of which the bid timing data were available for 171 (63.81%). The dummy is auction-specific and equal to 1 if the winning bid was received before the runner-ups, 0 otherwise. The time difference represents the time difference between the receival time of the winning bid and the receival time of the runner-up bid (which is negative if the winning bid was received first).

Table A14. Bidder distribution in 171 auctions examined in timing.

		Pol. conn. count	Pol. conn. share	Donators count	Donators share	Sus. firms count	Sus. firms share
Winners	Dummy = 0	98	0.57	142	0.83	156	0.91
	Dummy = 1	73	0.43	29	0.17	15	0.09
	No dummy	171	1	171	1	171	1
Runner-ups	Dummy = 0	90	0.53	153	0.89	153	0.89
	Dummy = 1	81	0.47	18	0.11	18	0.11
	No dummy	171	1	171	1	171	1
Both	Dummy = 0	188	0.55	295	0.86	309	0.90
	Dummy = 1	154	0.45	47	0.14	33	0.10
	No dummy	342	1	342	1	342	1

Notes: The dummy is auction-specific, and equal to 1 if the winning bid was received before the runner-ups, 0 otherwise.

Table A15. Robustness: adding covariates in close auction sample.

	Within 10% (main)					
	(1)	(2)	(3)	(4)	(5)	(6)
−5	0.132	0.013	0.013	0.013	0.013	0.013
	(0.612)	(0.842)	(0.716)	(0.716)	(0.593)	(0.572)
−4	0.337	0.064	0.064	0.064	0.064	0.064
	(0.612)	(0.842)	(0.716)	(0.716)	(0.593)	(0.572)
−3	0.362	0.106	0.106	0.106	0.106	0.106
	(0.612)	(0.842)	(0.716)	(0.716)	(0.593)	(0.572)
−2	0.627	0.454	0.454	0.453	0.453	0.453
	(0.612)	(0.842)	(0.716)	(0.716)	(0.593)	(0.572)
−1	0.743	0.441	0.441	0.440	0.440	0.440
	(0.612)	(0.842)	(0.716)	(0.716)	(0.593)	(0.572)
0	1.037*	1.013	1.013	1.012	1.012*	1.012*
	(0.612)	(0.842)	(0.716)	(0.716)	(0.593)	(0.572)
1	1.388**	1.455*	1.455**	1.454**	1.454**	1.454**
	(0.612)	(0.842)	(0.716)	(0.716)	(0.593)	(0.572)
2	1.476**	1.662**	1.662**	1.661**	1.661***	1.661***

(continued)

Table A15. (continued)

	Within 10% (main)					
	(1)	(2)	(3)	(4)	(5)	(6)
	(0.612)	(0.842)	(0.716)	(0.716)	(0.593)	(0.572)
3	1.856***	2.334***	2.334***	2.333***	2.333***	2.333***
	(0.612)	(0.842)	(0.716)	(0.716)	(0.593)	(0.572)
4	2.091***	2.570***	2.570***	2.569***	2.569***	2.569***
	(0.612)	(0.842)	(0.716)	(0.716)	(0.593)	(0.572)
5	2.298***	2.703***	2.703***	2.703***	2.703***	2.703***
	(0.612)	(0.842)	(0.716)	(0.716)	(0.593)	(0.572)
Political conn.	No	Yes	Yes	Yes	Yes	Yes
Log start emp.	No	No	Yes	Yes	Yes	Yes
Win margin	No	No	No	Yes	Yes	Yes
Firm FE	No	No	No	No	Yes	Yes
Auction FE	No	No	No	No	No	Yes
Mean employees	124.2909	124.2909	124.2909	124.2909	124.2909	124.2909
N	31,872	31,872	31,872	31,872	31,872	31,872
R^2	0.010	0.011	0.314	0.315	0.538	0.629
Adjusted R^2	0.009	0.009	0.284	0.285	0.509	0.544
Residual Std. Error	11.146 (df = 31,848)	11.144 (df = 31,824)	9.474 (df = 30,519)	9.469 (df = 30,507)	7.844 (df = 29,968)	7.563 (df = 25,934)

Notes: The main model, in Column (6), is estimated with Equation (1). The estimates are calculated using the package ("lfe," Gaure 2013) and show the LATE, difference in employment growth rates between winners and runner-ups. The point estimates and standard errors are transformed to absolute employment increases based on the coefficients and the mean number of employees (given in "mean employees") in the −sixth fortnight.
***, **, and * Significant at the 1%, 5%, and 10% level.

Table A16. Complaints—occurrence and distribution.

Complaint distribution				Complaint distribution		CPV's within our database	
Procuring entity	No.	Share	Four-digit CPV code	No.	Share	No.	Share
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Grad Zagreb	613	0.04	4523	662	0.32	1314	0.4
Hrvatske ceste d.o.o.	391	0.03	4521	205	0.1	422	0.13
HAC d.o.o.	316	0.02	4545	137	0.07	304	0.09
Hrvatske vode	316	0.02	4500	257	0.12	209	0.06
ZG holding d.o.o.	290	0.02	4526	104	0.05	198	0.06
HEP-ODS d.o.o.	284	0.02	4524	137	0.07	144	0.04
Hrvatske šume d.o.o.	248	0.02	4531	115	0.06	120	0.04
HŽ-Infrast. d.o.o.	244	0.02	4522	97	0.05	121	0.04
KBC Zagreb	239	0.02	4520	48	0.02	98	0.03
HP d.d.	225	0.01	4511	74	0.04	82	0.03
In top 10	3161	0.22	In top 10	1836	0.89	3012	0.92
Total	15,230	1	Total	2063	1	3257	1

Notes: The table shows the top 10 procuring entities that received the most complaints, as well as the top 10 four-digit CPV codes with the most complaints. Columns (5) and (6) show the occurrence of complaints through the four-digit CPV codes. Columns (7) and (8) show the CPV distribution through our non-filtered database of auctions. CPVs are ordered by Column (8).

Table A17. Overview of all donations to political parties.

	Donations							
	To all parties (1)	**To ruling parties (2)**	**No. per party (3)**	**Val. per party (4)**	**No. per donator (5)**	**Val. per donator (6)**	**No. per donator (in sample) (7)**	**Val. per donator (in sample) (8)**
Sum	2,937,695.86	1,931,103.91	3957	2,937,695.86	3957	2,937,695.86	108	341,382.59
Mean	742.4	921.33	63.82	47,382.19	1.17	866.83	1.57	4947.57
Std. dev.	1522.58	1868.65	269.59	246,989.47	0.74	2117.85	0.88	7459.83
10th	26.67	66.67	1	340	1	26.67	1	400
50th	266.67	266.67	6.5	1897.53	1	266.67	1	1733.33
90th	1992	2400	98.2	44,251.93	2	2266.67	3	14,133.33
Max	26,666.67	26,666.67	2095	1,929,770.58	36	53,333.33	4	34,786.67
Observation	3957	2096	62	62	3389	3389	69	69

Notes: The first column shows info on all 3957 donations to any party preceding the parliamentary elections in 2016 and those in 2017, while the second column shows donations to the ruling party after the election (HDZ). Columns (3) and (4) examine donations by the party which they target. Columns (5) and (6) do the same but instead by the donation origin (568 donations had no identification number connected to them but none of them were donations by firms, those are excluded, hence the lower observation number). Columns (7) and (8) look at only the donations given by construction firms within our sample of PPCs. All monetary values are in euro.

Table A18. MEAT criteria distribution.

Price crit. (1)	No. (2)	Share (3)	Cost crit. (4)	No. (5)	Share (6)	Quality crit. (7)	No. (8)	Share (9)	Other crit. (10)	No. (11)	Share (12)
90	619	0.61	>30	7	0.01	>30	30	0.03	>10	3	0.00
80–89	285	0.28	11–30	90	0.09	11–30	281	0.28	10	2	0.00
70–79	71	0.07	10	91	0.09	10	551	0.54	5	1	0.00
<70	46	0.05	0–9	833	0.82	0–9	159	0.16	0	1015	0.99

Notes: The table shows MEAT criteria distribution of auctions for which we have the criteria data. The observed dataset contains 3493 bids across 1101 auctions which were awarded via the MEAT criteria. Of those 1101 auctions, we have the criteria data for 1021 of them, for which the distribution is shown above.

Table A19. Auction criteria characteristics.

Sample (1)	Variable (2)	Sum (3)	Mean (4)	Std. dev. (5)	10th (6)	50th (7)	90th (8)	Obs. (9)	Raw obs. (10)
LP and MEAT	Winning bid	2062.55	0.72	6.47	0.06	0.18	0.88	2859	3257
	Bid of runner-up	2362.39	0.83	7.67	0.07	0.21	1.01	2859	3257
	Est. value	2310.79	0.81	5.96	0.08	0.22	1.07	2859	3257
	Number of bids		4	2.01	2	3	7	10,232	11,873
	Number of auctions							2859	3257
	Winning bid	1276.89	0.73	4.8	0.06	0.18	0.95	1758	1983
LP	Bid of runner-up	1368.14	0.78	4.94	0.07	0.2	1.01	1758	1983
	Est. value	1163.28	0.83	5.02	0.08	0.21	1.07	1758	1983
	Number of bids		4.16	2.14	2	4	7	6739	7743
	Number of auctions							1758	1983
	Winning bid	785.66	0.71	8.48	0.05	0.18	0.79	1101	1274
	Bid of runner-up	994.25	0.9	10.68	0.07	0.24	1.01	1101	1274
MEAT	Est. value	856.69	0.78	7.21	0.08	0.24	1	1101	1274
	Number of bids		3.74	1.76	2	3	6	3493	4130
	Number of auctions							1101	1274

Notes: The table shows auction data characteristics across auctions awarded via LP and MEAT, separately and when grouped together, after the exclusion. The last two columns show the observations, the last column shows the observations before the exclusion of the bidders for which we do not have the necessary financial and employment data for the analysis, and the Obs. column shows the observations after the exclusion. All monetary values are in mil. Euro. VAT is not included.

Table A20. Contamination of evaluation period and seasonality.

	Main (within 10%) (1)	In-season (2)	Off-season (3)	One auction victors (4)	+ window (5)	+ month (6)
−5	0.013	0.162	−0.135	−0.479	0.013	0.013
	(0.572)	(0.827)	(0.790)	(0.968)	(0.572)	(0.572)
−4	0.064	−0.163	0.352	−0.328	0.064	0.064
	(0.572)	(0.827)	(0.790)	(0.968)	(0.572)	(0.572)
−3	0.106	−0.186	0.467	−0.093	0.106	0.106
	(0.572)	(0.827)	(0.790)	(0.968)	(0.572)	(0.572)
−2	0.453	0.078	0.901	0.398	0.453	0.453
	(0.572)	(0.827)	(0.790)	(0.968)	(0.572)	(0.572)
−1	0.440	0.175	0.766	0.581	0.440	0.440
	(0.572)	(0.827)	(0.790)	(0.968)	(0.572)	(0.572)
0	1.012*	0.499	1.622**	1.421	1.012*	1.012*
	(0.572)	(0.827)	(0.790)	(0.968)	(0.572)	(0.572)
1	1.454***	1.095	1.895**	2.351**	1.454**	1.454***
	(0.572)	(0.827)	(0.790)	(0.968)	(0.572)	(0.572)
2	1.661***	1.684**	1.668**	2.820***	1.661***	1.661***
	(0.572)	(0.827)	(0.790)	(0.968)	(0.572)	(0.572)
3	2.333***	2.633***	2.004***	3.323***	2.333***	2.333***
	(0.572)	(0.827)	(0.790)	(0.968)	(0.572)	(0.572)
4	2.569***	3.010***	2.083***	3.492***	2.569***	2.569***
	(0.572)	(0.827)	(0.790)	(0.968)	(0.572)	(0.572)
5	2.703***	3.262***	2.064***	4.113***	2.703***	2.703***
	(0.572)	(0.827)	(0.790)	(0.968)	(0.572)	(0.572)
Mean employees	124.2909	126.7562	121.3892	85.7674	124.2909	124.2909
N	31,872	17,232	14,640	7,524	31,872	31,872
R^2	0.629	0.635	0.623	0.598	0.629	0.629
Adjusted R^2	0.544	0.545	0.528	0.491	0.544	0.544
Residual Std. Error	7.563 (df = 25,934)	8.066 (df = 13,834)	7.060 (df = 11,689)	6.771 (df = 5937)	7.563 (df = 25,934)	7.563 (df = 25,934)

Notes: Columns (2) and (3) split the sample in two parts, Column (2) contains each auction awarded from April to (and including) October, Column (3) contains the rest. Column (4) examines the effect on the victors whose only winning bid during the following three months is from the observed auction. The final two columns include the value a bidder won during three months (in Column (5)), and the month in which the auction was awarded (Column (6)) as additional fixed effects. The dependent variable is employment growth at firm–auction level in each fortnight period (from −6 to 5). The model is estimated with Equation (1). The independent variables are the fortnight periods, win margin-, auction-, and firm-fixed effects, political connection dummy, as well as the (log) number of firms' employees in the −sixth fortnight. The estimates are calculated using the package ("lfe," Gaure 2013) and show the LATE, difference in employment growth rates between winners and runner-ups. The point estimates and standard errors are transformed to absolute employment increases based on the coefficients and the mean number of employees (given in "mean employees") in the −sixth fortnight.
***, **, and * Significant at the 1%, 5%, and 10% level.

Table A21. Characteristics of winners' new employees: education level, sources of previous employment, and mean age.

Education level	Previous employment	Number of new employees	Mean employee age
Higher educated	Different firm	240	38.56
	No previous employment	538	32.89
	Same firm	488	35.80
	Total	1266	35.08
Lower educated	Different firm	1332	38.49
	No previous employment	3976	34.62
	Same firm	4711	38.65
	Total	10,019	37.03
Any education level	Different firm	1572	38.50
	No previous employment	4514	34.41
	Same firm	5199	38.38
	Total	11,285	36.81

Table A22. Professions of the winners' new employees.

NKD code	Number of new employees
(7122) Masons	996
(7124) Carpenters and joiners	799
(9911) Workers without occupations	774
(9312) Civil engineering workers	638
(8332) Operators of construction, and similar machinery	634
(8324) Drivers of heavy goods vehicles and towing vehicles	602
(3112) Architectural, civil and geodetic engineers, and technicians	571
(7222) Toolmakers and related occupations	438
(7129) Other masonry occupations	389
(9132) Cleaners and maids	361
In top 10 professions	6202
Total	11,285

Table A23. The sector where the winners' unique new employees were previously employed: subsample of employees coming from different firms.

Sector of previous employment	Number of new employees
F—Construction	1145
C—Manufacturing	146
G—Wholesale and retail trade; repair of motor vehicles and motorcycles	110
M—Professional, scientific, and technical activities	55
N—Administrative and support service activities	40
H—Transportation and storage	27
E—Water supply, sewerage, waste management, and remediation activities	14
B—Mining and quarrying	10
L—Real-estate activities	10
I—Accommodation and food service activities	4
D—Electricity, gas, steam, and air conditioning supply	3
J—Information and communication	3
A—Agriculture, forestry, and fishing	2
R—Arts, entertainment, and recreation	2
S—Other service activities	1
Total	1572

Table A24. Quantification.

		Auction awarded value		Quantification of a single employee		
		Median	**Mean**	**Employee effect**	**Cost (by median)**	**Cost (by mean)**
Entire sample		237,720.50	941,227.90	2.55	93,223.73	369,108.98
	10%	268,586.67	1,024,784.20	2.70	99,476.54	379,549.70
By bid "closeness"	.5–4%	281,717.33	772,719.95	3.00	93,905.78	257,573.32
	2nd bin	220,756.67	452,221.20	2.20	100,343.94	205,555.09
By outsourcing	1st bin	213,974.67	1,119,671.74	3.65	58,623.20	306,759.38
	<2.6%	146,046.58	236,127.99	0.09	1,622,739.81	2,623,644.37
	2.6–7.3%	203,085.00	433,268.21	1.60	126,928.12	270,792.63
By relative	7.3–19.8%	313,417.17	617,911.48	2.35	133,369.01	262,941.05
auction size	>19.8%	641,412.00	2,922,715.31	3.47	184,844.96	842,281.07

Notes: All monetary values are given in euros. VAT is included.

APPENDIX D: FIGURES

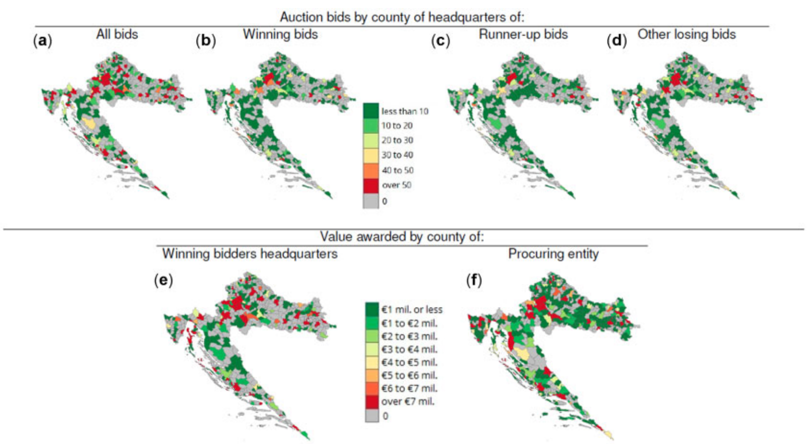

Figure A1. PPC value and auctions awarded in auctions with multiple bidders.

Notes: VAT is included. (**a**) Number of all bids by bidder municipality, (**b**) Number of winning bids by winner municipality, (**c**) Number of runner-up bids by runner-up municipality, (**d**) Number of other losing bids by bidder municipality, (**e**) Awarded value by municipality of winning bidder and (**f**) Awarded value by municipality of procuring entity.

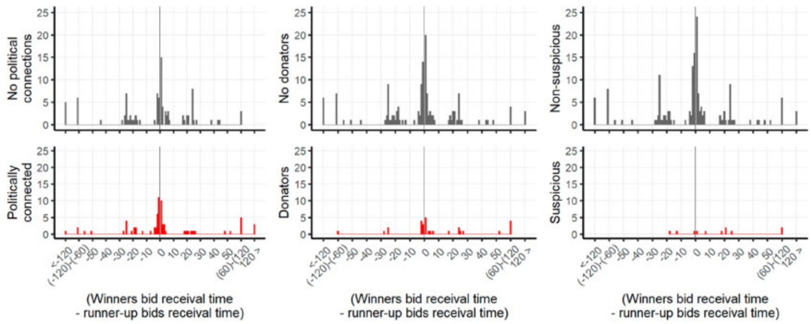

Figure A2. Histograms for winner distribution by timing (in hours).

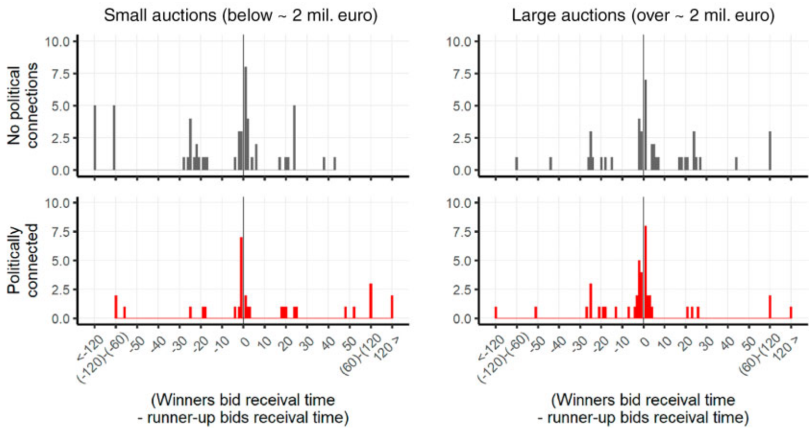

Figure A3. Histograms for timing by political connection distribution by auctions size (in hours).

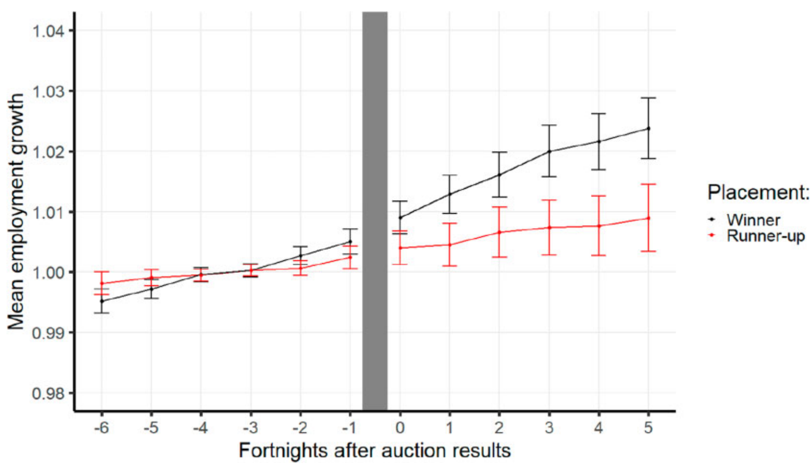

Figure A4. Mean employment before and after receiving a PPC (within 10% close auctions).

Notes: Unconditional means are given in the figure. The day-by-day data on firms' number of employees are shown relative to the average number of employees during the sixth fortnight prior to the auction results. That gives us the employment growth rate for each firm. We then aggregate (average) the day-by-day growth rate to fortnights (14-day periods). Confidence intervals are given at 90%.

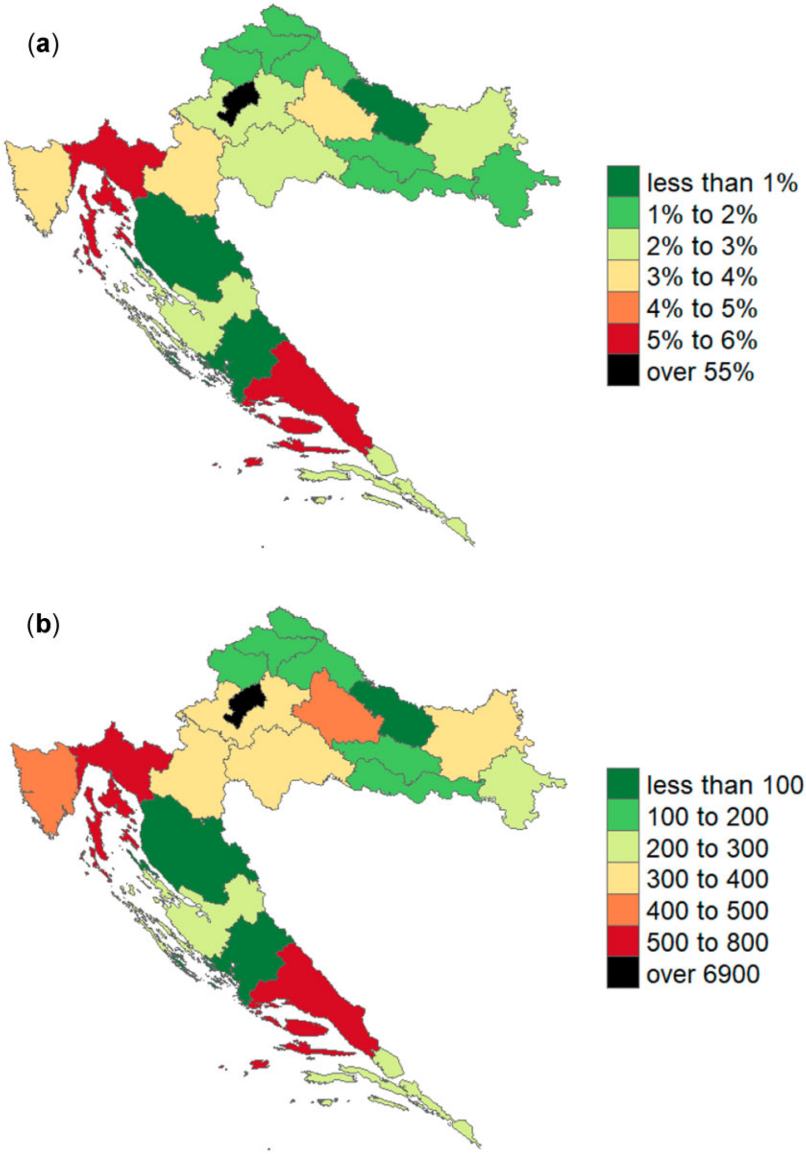

Figure A5. Complaints by county.

Notes: The data encompasses the entire database of complaints. It shows the distribution of fewer than 15,230 complaints by county of the procuring entity (for which we have the data on their location). (**a**) The percentage of complaints and (**b**) the total number of complaints by county of procuring entity.

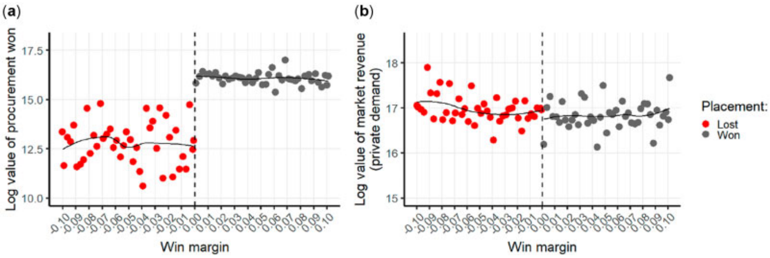

Figure A6. PPC value won versus market value acquired.

Notes: The *X*-axis represents the "win margin" of a bid, it is essentially the criteria we use for defining "closeness" in an auction (We refer to the Table A2), only we multiply it by −1 if the bid is a losing one. The *Y*-axis represents the natural log of the total procurement value won by a firm, graph (**a**), and graph (**b**) the natural log of the firm's revenue in the examined year. Points represent bins formed according to the winning margin (sizes of 0.0025).

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